

Stopped Experiments: From Bayes-Risk Contraction to Minimax Regret

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

Abstract

We characterize finite-horizon regret through the cost of reducing posterior Bayes risk by a fixed factor. At scale b , a stopped experiment ends on contraction or abandonment and costs b times expected regret divided by its contraction probability. Balancing the capped cost $SC_{b,T}/b$ against commitment bT over geometric risk shells gives an envelope that lower-bounds worst-prior Bayes regret and upper-bounds it within $O(\log T)$. Under measurable selection and Bayes-minimax equality, the same envelope characterizes minimax regret for finite latent and action classes and for Gaussian-feedback loss classes on compact action spaces.

For paired contextual top- k bandits with aggregate Gaussian feedback, every finite sign class $\mathcal{F}$ has stopped cost $O(\log(e|\mathcal{F}|))$, and balanced product classes attain this dependence. For $M \geq 2^k$, maximizing over known signal scales and classes of cardinality at most M , the aggregate-feedback value is $\Theta(k \min\{T, \sqrt{T \log M}\})$ in selected-sum units. On those balanced product classes, semi-bandit feedback lowers stopped cost by a factor k . This gives a factor- k separation on the same horizon-calibrated, aggregate-hard class and a factor- $\sqrt{k}$ separation when each feedback model is tuned to its own hardest signal scale. The one-context construction, augmented with known mean-zero decoys, matches the known upper bound up to logarithms throughout the constant-density regime. For finite linear preference classes, we calculate the exact dependence on binary-channel capacity and evaluator sensitivity. On a dimple family, unrestricted observation-adaptive experiments have stopped cost $\tilde{O}(d^2)$, whereas adaptive blocks in fresh random subspaces incur a polynomially larger cost.

1 Introduction

How much regret must be spent before observations make a decision problem meaningfully easier? A fixed-confidence guarantee may count the observations needed to identify a good action, but it need not control the regret incurred by those queries or a policy that learns only on favorable transcript branches. Nor is a single query always the right unit: an early observation may matter chiefly because it determines which later queries should be made. We therefore study complete adaptive experiments whose stopping target is a reduction in posterior decision risk.

Stopped risk-contraction experiments. We begin with a fixed latent environment ω . An action a incurs a nonnegative decision gap $\Delta_\omega(a)$ and produces an observation. If Q_t is the posterior after t observations, its Bayes decision risk is

$$r(Q_t) = \inf_a \mathbb{E}_{Q_t} \Delta_\omega(a).$$

At a state where this risk is of order b , an adaptive experiment runs until it is abandoned or first reaches $r(Q_t) \leq \gamma b$. If N is its duration and $\mathbf{M}$ the first-hit event, we define the pointwise stopped

cost by

$$\text{SC}_b(Q) = \inf_{\text{stopped experiments}} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\text{M})}. \quad (1)$$

We call this a stopped risk-contraction experiment, or simply a stopped experiment. Thus $\text{SC}_b(Q)$ is the regret paid per unit probability of a decision-relevant reduction in risk, after scaling by the current risk level. The definition introduces no auxiliary information budget or dimension-dependent normalization. A separate query charge is unnecessary: every query before the hit has posterior expected gap at least γb , so $\mathbb{E} R_N \geq \gamma b \mathbb{E} N$.

Two-sided regret transfer. The paper’s central result transfers between stopped risk contraction at one posterior and scale and finite-horizon regret. For an upper bound, one constructs at each risk scale an adaptive experiment that contracts posterior risk at controlled regret cost; the theorem concatenates these designs across scales. For a lower bound, one strengthens a testing or estimation inequality so that it applies to stopped transcripts and remains linear in the probability of contraction; the theorem then balances the branch that contracts with the branch that continues to incur risk. The transfer is not automatic. Ordinary sample complexity neither prices the decisions used to collect data nor controls policies that learn only on favorable transcript branches.

Let $\text{SC}_{b,T}$ restrict the experiment to at most T queries. On geometric risk shells, define the horizon-matched envelope by balancing

$$\text{commitment } bT \quad \text{against} \quad \text{experimentation } \text{SC}_{b,T}/b$$

and taking the largest matched value over the scales $b \geq T^{-1}$. Let Ψ_T^{pt} optimize the experiment separately at each posterior and let Ψ_T^{unif} require a measurable state-dependent rule. The central theorem proves

$$c_\gamma \Psi_T^{\text{pt}} \leq \mathfrak{R}^{\text{B}}(d, T) \leq 1 + C_\gamma (\log T) \Psi_T^{\text{unif}}.$$

The lower proof stops an arbitrary regret policy at its first contraction and balances the hitting and non-hitting branches. The upper proof chooses between commitment and a cap- T experiment at each scale, then composes the chosen experiments using posterior-risk supermartingality.

For finite latent and action classes, and for Gaussian-feedback loss classes on compact action spaces, statewise experiments are measurably selectable and worst-prior Bayes regret equals minimax regret. The same envelope therefore characterizes minimax regret at the same horizon, within the logarithmic factor. The exponent identity

$$\alpha^{\text{M}} = \alpha^{\text{B}} = \frac{1}{2} \alpha^{\text{stop}}$$

is a coarser consequence.

The results below play different roles. The finite-horizon envelope is the general characterization. The contextual top- k theorem gives a sharp finite-class calculation and isolates the statistical price of aggregate feedback. The dimple construction gives a structural separation robust to adaptive blocks in fresh random subspaces. The preference theorem determines how binary comparison capacity and evaluator sensitivity enter the stopped cost uniformly over a broad linear action-set sandwich.

1. For contextual top- k selection with $2k$ independent Gaussian arms and aggregate feedback, let $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \{-1, +1\}^k\}$ be a finite realizable sign class. We prove the uniform certificate

$$\sup_{(Q,b) \text{ active}} \text{SC}_b^{\text{agg}}(Q) \lesssim \log(e|\mathcal{F}|).$$

For q balanced contexts and the full product class, $\log_2 |\mathcal{F}| = qk$, this bound is attained:

$$\text{SC}_\varepsilon^{\text{agg}}(Q_0) = \Theta_\gamma(qk), \quad \text{SC}_\varepsilon^{\text{semi}}(Q_0) = \Theta_\gamma(q).$$

Consequently, for $M \geq 2^k$, maximizing over $|\mathcal{F}| \leq M$ and the known signal scale, the aggregate-feedback value is $\Theta(k \min\{T, \sqrt{T \log M}\})$. On balanced product classes, semi-bandit feedback is better by a factor k on the same horizon-calibrated aggregate-hard class and by $\sqrt{k}$ when each feedback model is tuned separately. With one context this recovers $\tilde{\Theta}(\min\{kT, k^{3/2}\sqrt{T}\})$, closing the dense independent-Gaussian slice of the gap in [Rejwan and Mansour \(2020\)](#). Adding known mean-zero decoys extends the lower bound to all $n \geq 2k$, and hence matches the known upper bound up to logarithms throughout the constant-density regime $2k \leq n \leq Ck$.

2. For every compact linear action set satisfying $\mathcal{H}_d \subseteq \mathcal{X} \subseteq B_2^d(1)$, under probit or Bradley–Terry comparisons of the hypercube parameters, we prove

$$\text{SC}_b^{\text{pref}}(Q_0) = \Theta_{\gamma, F} \left(b^2 d + \frac{d^2}{\beta^2} \right).$$

The two terms are a one-bit capacity floor and a weak-signal cost. The result gives fixed-scale regret up to a logarithm and an unknown-scale envelope, and it applies both to deterministic trajectory preferences and to the full dense top- d hypersimplex. Every comparison charges the average loss of its two actions; total two-action regret is exactly twice the displayed convention.

3. For the full BCO class, the closure theorem restates the current gap as $5/2 \leq \alpha_{\text{stop}} \leq 3$. On the dimple family of [Bakhtiari et al. \(2025\)](#), an unrestricted observation-adaptive experiment has stopped cost $\tilde{O}(d^2)$. If an experiment may adapt for at most ℓ queries inside each fresh random m -subspace but cannot orient later subspaces from earlier data, then, at $\varepsilon = m/d$,

$$\text{SC}_b^{\text{bf}(m, \ell)} \geq c_\gamma \frac{d^3}{\ell m^2}.$$

This yields a polynomial finite-horizon regret separation whenever $\ell m^2 \leq d^{1-\delta}$. It separates experiment classes; it is not an unrestricted lower bound for the family.

The results require nonnegative additive gaps, a posterior that is sufficient for future decisions, and repetition of the same latent decision problem after each observation. Convexity and Gaussian observations are needed only in the measurable-selection and BCO sections.

1.1 Related work

Regret complexity and sequential design. Information ratios, AIR, and DEC turn one-step regret–information or decision–estimation comparisons into sequential regret bounds ([Russo and Van Roy, 2016, 2018](#); [Lattimore and György, 2021](#); [Foster et al., 2021, 2022, 2023](#); [Xu and Zeevi, 2025](#); [Neu et al., 2024](#)); chained ratios organize such comparisons across action scales ([Gouverneur et al., 2024](#)). These methods may re-optimize after every observation. The difference is therefore not whether the policy is adaptive, but the primitive certificate: stopped cost evaluates an entire adaptive experiment, including how early observations determine later queries and stopping, rather than assigning a value to one query at a time. Interactive Le Cam, Fano, and Assouad inequalities provide lower-bound tools for the resulting adaptive transcript laws ([Chen et al., 2024](#)). The construction is also related to sequential experimental design. Blackwell comparison and Bayesian theories of experiments value observations through downstream decisions ([Blackwell, 1953](#); [Lindley,](#)

1956; DeGroot, 1962), while sequential decision theory and active testing choose experiments and stopping rules (Arrow et al., 1949; Chernoff, 1959; Naghshvar and Javidi, 2013). Those theories motivate the experiment language, but their standard sample or terminal-decision costs are not used as black-box inputs here. The required certificate is more specific: it charges the decision loss of every query, stops at a fixed-factor reduction in posterior Bayes risk, and retains the probability of reaching that set.

Risk contraction and adaptive experiments. Several approaches avoid exact identification by seeking only enough information to choose an action whose loss is within a prescribed tolerance of optimal (Dong and Van Roy, 2018; Russo and Van Roy, 2022; Arumugam and Van Roy, 2021; Bubeck and Sellke, 2023). This is the relevant meaning of a near-optimal target: at tolerance b , one need not identify the environment or a unique optimizer. Here the target is posterior—success means $r(Q_N) \leq \gamma b$ —and different transcripts may end with different near-optimal actions. Related pointwise Gaussian analysis localizes complexity around an individual target rather than the full ambient class (Xu, 2026b). In this paper, “pointwise” has the narrower technical meaning that the experiment is optimized separately at each posterior state; measurable selection asks whether these statewise choices can be implemented by one rule. Finally, Maiti et al. (2026) show in linear best-arm identification that early observations can reveal which later probes are useful. The present paper uses neither result as a technical premise: they clarify, respectively, why the relevant target is decision-level risk rather than exact model recovery and why a complete adaptive experiment can be strictly cheaper than one-query certificates.

Closest application precedents. The Bellman-sufficient framework of Xu (2026a) accommodates physical states, contexts, and other history-dependent variables. The present theory treats the repeated latent-decision specialization in which the posterior is sufficient and the same nonnegative gap recurs; these restrictions make posterior risk persistent enough for the two-sided transfer. On posterior space, the stopped experiment is a stochastic shortest-path problem (Bertsekas and Tsitsiklis, 1991). Application-specific comparisons accompany the corresponding results. Table 1 shows that the top- k theorem closes the constant-density, independent-Gaussian slice of the gap displayed by Rejwan and Mansour (2020); the contextual extension identifies the exact worst finite-class dependence and separates aggregate from semi-bandit feedback. The dependence on n when n/k diverges remains open. Contextual combinatorial full-bandit and semi-bandit precedents are compared directly with the paired finite-class result in Section 3.1 (Zierahn et al., 2023; Erez and Koren, 2025; Qin and Zhang, 2026). The preference calculation extends the paired hypercube core through a binary comparison channel and uniformly enlarges the action set up to the Euclidean ball. General linear trajectory-preference regret is studied by Saha et al. (2023); the present result isolates the exact stopped cost rather than claiming a new general tabular-RL rate (Xu et al., 2020; Wang et al., 2023). For BCO, the main model-specific conclusion is the polynomial separation, on the dimple family of Bakhtiari et al. (2025), between unrestricted observation-adaptive experiments and adaptive blocks confined to fresh random subspaces.

2 Stopped risk contraction and regret transfer

2.1 The repeated latent decision problem

We first study a repeated decision problem with a fixed latent environment; convexity is not required. This is the static-state specialization of the Bellman-sufficient framework of Xu (2026a): the current

posterior is the pre-action state, the same action problem recurs after every observation, and the action gap is a nonnegative one-step Bellman loss. General stateful control may also possess a Bellman-sufficient state, but need not satisfy the persistence property used below to concatenate experiments across risk scales.

Let Ω , $\mathcal{A}$, and $\mathcal{O}$ be standard Borel environment, action, and observation spaces, and let $\Delta_\omega(a) \in [0, 1]$ be a nonnegative action gap. Querying a produces an observation in $\mathcal{O}$ from a kernel $P_\omega(\cdot | a)$. The environment $\omega \sim Q$ is drawn once and remains fixed. A policy chooses A_t from the pre-action history H_{t-1} , observes O_t , and incurs $\Delta_\omega(A_t)$. We include all independent exogenous policy randomization in H_0 and write $\mathcal{H}_t = \sigma(H_t)$ for the completed history filtration.

For a posterior Q , define the Bayes risk

$$r(Q) = \inf_{a \in \mathcal{A}} \mathbb{E}_Q \Delta_\omega(a).$$

We assume the posterior is a sufficient controlled state, r is measurable, and measurable ϵ -Bayes actions exist. These are standard in compact Borel models and are verified below for Gaussian-feedback loss classes on compact action spaces. The theorem applies to static latent-model decision problems of the DMSO type (Foster et al., 2021). Neither the action space nor the map $a \mapsto \Delta_\omega(a)$ is assumed to be convex. Nonnegativity is essential. If one-step increments can be signed, stopping may discard later negative increments that would offset earlier positive ones, and regret need not control the number of elapsed queries. Such problems require a separate query charge or a lower-bounded Bellman loss.

The repeated-decision structure implies that posterior Bayes risk is a nonnegative supermartingale. Indeed,

$$\mathbb{E}[r(Q_{t+1}) | \mathcal{H}_t] \leq \inf_{a \in \mathcal{A}} \mathbb{E}[\mathbb{E}\{\Delta_\omega(a) | \mathcal{H}_{t+1}\} | \mathcal{H}_t] = r(Q_t). \quad (2)$$

This fact, rather than convexity or a particular observation law, controls returns to a risk scale in the upper bound.

Fix

$$0 < \gamma < \vartheta < 1. \quad (3)$$

A pair $s = (Q, b)$ is *active* when

$$\vartheta b < r(Q) \leq b.$$

Starting from an active state, let

$$\tau_b = \inf\{t \geq 1 : r(Q_t) \leq \gamma b\} \quad (4)$$

be the first posterior-risk contraction.

Definition 2.1 (Stopped cost). *An admissible experiment consists of an adaptive query policy and a positive, possibly infinite, $(\mathcal{H}_t)$ -stopping time σ . It runs for*

$$N = \sigma \wedge \tau_b$$

rounds and succeeds on $\mathbf{M} = \{\tau_b \leq \sigma\}$. We require $N < \infty$ almost surely. Expectations below are initially extended-valued; Lemma 2.2 shows that every experiment with finite displayed ratio has $\mathbb{E}N < \infty$. Define

$$\text{SC}_b(Q) := \inf_{\pi, \sigma} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}, \quad R_N = \sum_{t=1}^N \Delta_\omega(A_t), \quad (5)$$

with ratio $+\infty$ when the success probability is zero.

The first-hit convention loses nothing: after reaching the lower-risk set, additional observations cannot improve the success event and only add nonnegative regret. It also ensures that every charged query is made while risk exceeds γb , which is what turns elapsed queries into regret. Voluntary stopping is essential because an adaptive design may abandon an unpromising branch. Dividing by $\mathbb{P}_Q(\mathbf{M})$ prevents rare successful branches from making such abandonment artificially free.

Lemma 2.2 (Query cost is already charged). *Every admissible experiment satisfies*

$$\mathbb{E}_Q R_N \geq \gamma b \mathbb{E}_Q N. \quad (6)$$

Consequently,

$$\text{SC}_b(Q) \leq \inf_{\pi, \sigma} \frac{b^2 \mathbb{E}_Q N + b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})} \leq (1 + \gamma^{-1}) \text{SC}_b(Q).$$

The proof is a stopped-sum application of the defining inequality $r(Q_{t-1}) > \gamma b$ before the first hit; see Appendix A.

2.2 Bellman representation of the stopped ratio

The stopped cost is a stochastic shortest-path problem on posterior space. For a query cap L and price $q > 0$, put $W_0^q = 0$ and

$$\begin{aligned} J_k^q(Q, b) &= \inf_{a \in \mathcal{A}} \left\{ b \mathbb{E}_Q \Delta_\omega(a) \right. \\ &\quad \left. + \int [-q \mathbf{1}\{r(Q^{a,o}) \leq \gamma b\} + \mathbf{1}\{r(Q^{a,o}) > \gamma b\} W_{k-1}^q(Q^{a,o}, b)] P_Q(do | a) \right\}, \quad (7) \\ W_k^q(Q, b) &= \min\{0, J_k^q(Q, b)\}. \end{aligned}$$

The root uses J_L^q , so the experiment makes at least one query; W_k^q permits voluntary stopping on a non-hitting branch. If $\text{SC}_{b,L}$ is the infimum in (5) over experiments with $N \leq L$ almost surely, then

$$\text{SC}_{b,L}(Q) < q \iff J_L^q(Q, b) < 0, \quad \text{SC}_b(Q) = \inf_{L \geq 1} \text{SC}_{b,L}(Q). \quad (8)$$

The first equivalence follows by minimizing $b \mathbb{E} R_N - q \mathbb{P}(\mathbf{M})$; the second follows by truncating the stopping time. The continuation value at $Q^{a,o}$ credits an early observation for improving later queries, even when that observation does not itself reach the contraction set.

2.3 Pointwise and uniformly selectable stopped-cost profiles

Let $\mathfrak{D} = (\mathfrak{D}_d)$ be a sequence of such decision problems. For a scale window $B^{-1} \leq b \leq 1$, write $\mathbf{S}_{\mathfrak{D}}(d, B)$ for the active posterior states. Define

$$\text{SC}_{\mathfrak{D}}^{\text{pt}}(d, B) = \sup_{(Q,b) \in \mathbf{S}_{\mathfrak{D}}(d,B)} \inf_{\varepsilon} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}, \quad (9)$$

$$\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, B) = \inf_{\Sigma \text{ universally measurable}} \sup_{(Q,b) \in \mathbf{S}_{\mathfrak{D}}(d,B)} \frac{b \mathbb{E}_Q^{\Sigma(Q,b)} R_N}{\mathbb{P}_Q^{\Sigma(Q,b)}(\mathbf{M})}. \quad (10)$$

The supremum of an empty set is zero. Always $\text{SC}_{\mathfrak{D}}^{\text{pt}} \leq \text{SC}_{\mathfrak{D}}^{\text{unif}}$. The first quantity chooses the best experiment separately at each state. The second requires all statewise choices to form one universally measurable experiment rule.

For a finite-horizon statement, it is useful to retain both the risk scale and the experiment cap. Put $b_j = \vartheta^j$, and let $\mathcal{S}_j(d)$ be the posterior states Q satisfying $\vartheta b_j < r(Q) \leq b_j$. Recalling that $\text{SC}_{b,T}$ restricts the experiment to $N \leq T$, define

$$\mathcal{C}_{j,T}^{\text{pt}}(d) = \sup_{Q \in \mathcal{S}_j(d)} \text{SC}_{b_j,T}(Q), \quad (11)$$

$$\mathcal{C}_{j,T}^{\text{unif}}(d) = \inf_{\substack{\Sigma_{j,T} \text{ universally measurable} \\ N^{\Sigma_{j,T}(Q)} \leq T}} \sup_{Q \in \mathcal{S}_j(d)} \frac{b_j \mathbb{E}_Q^{\Sigma_{j,T}(Q)} R_N}{\mathbb{P}_Q^{\Sigma_{j,T}(Q)}(\mathbf{M})}. \quad (12)$$

If the problem class contains several known action spaces, the states in these displays include the identity of the action space. For $T \geq 2$, let

$$J_T = \max\{j \geq 0 : b_j \geq 1/T\}$$

and introduce the horizon-matched stopped envelopes

$$\Psi_{\mathfrak{D}}^{\diamond}(d, T) = \max_{0 \leq j \leq J_T} \min \left\{ b_j T, \frac{\mathcal{C}_{j,T}^{\diamond}(d)}{b_j} \right\}, \quad \diamond \in \{\text{pt}, \text{unif}\}. \quad (13)$$

The first term is the cost of making no further experiment and repeatedly using a current Bayes action. The second is the regret per unit probability of contracting at that scale. Thus the envelope compares experimentation and commitment before maximizing over the scales relevant to horizon T .

Let

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) = \sup_Q \inf_{\pi} \mathbb{E}_Q^{\pi} R_T$$

denote worst-prior Bayes regret. If $\mathfrak{D}_d$ contains several known action spaces, the definition also takes the supremum over those spaces.

Theorem 2.3 (Stopped-cost regret transfer). *For every $T \geq 2$ and $T^{-1} \leq \eta < 1/2$,*

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq \min \left\{ T, C_{\gamma, \vartheta} \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T)}{\eta} + 2T\eta \right\}. \quad (14)$$

In particular,

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq C_{\gamma, \vartheta} \min \left\{ T, 1 + \sqrt{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) T} \right\}. \quad (15)$$

Conversely, for every active state (Q, b) , every $0 < c < \text{SC}_b(Q)$, and every $n \geq 1$,

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, n) \geq \inf_{\pi} \mathbb{E}_Q^{\pi} R_n \geq \frac{\gamma c b n}{c + \gamma b^2 n} \geq \frac{1}{2} \min \left\{ \gamma b n, \frac{c}{b} \right\}. \quad (16)$$

In particular, if $0 < c < \text{SC}_{\mathfrak{D}}^{\text{pt}}(d, B)$, some active state with $B^{-1} \leq b \leq 1$ satisfies this premise. If $c/(\gamma b^2) \geq 2$, then at

$$n = \left\lfloor \frac{c}{\gamma b^2} \right\rfloor$$

one has

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, n) \geq \frac{\sqrt{\gamma}}{4} \sqrt{c n}. \quad (17)$$

Theorem 2.4 (Finite-horizon stopped-envelope characterization). *Let*

$$\mathcal{E}_T = \left\{ 0 \leq j \leq J_T : \frac{\mathsf{C}_{j,T}^{\text{unif}}(d)}{b_j} \leq b_j T \right\}.$$

For every $T \geq 2$,

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \geq \frac{\gamma}{1+\gamma} \Psi_{\mathfrak{D}}^{\text{pt}}(d, T), \quad (18)$$

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq 1 + \frac{1}{1-\gamma/\vartheta} \sum_{j \in \mathcal{E}_T} \frac{\mathsf{C}_{j,T}^{\text{unif}}(d)}{b_j} + \max_{\substack{0 \leq j \leq J_T \\ j \notin \mathcal{E}_T}} b_j T \quad (19)$$

$$\leq 1 + \frac{1}{1-\gamma/\vartheta} \sum_{j=0}^{J_T} \min \left\{ b_j T, \frac{\mathsf{C}_{j,T}^{\text{unif}}(d)}{b_j} \right\} \quad (20)$$

$$\leq 1 + \frac{J_T + 1}{1-\gamma/\vartheta} \Psi_{\mathfrak{D}}^{\text{unif}}(d, T). \quad (21)$$

The maximum over an empty complement of $\{0, \dots, J_T\}$ in (19) is zero. In particular, the loss from replacing the full shellwise profile by its largest horizon-matched value is at most logarithmic in T .

Corollary 2.5 (Two-sided finite-horizon closure). *Suppose the cap- T statewise stopped experiments admit universally measurable ϵ -optimal choices and finite-horizon Bayes–minimax equality holds. Then*

$$\Psi_{\mathfrak{D}}^{\text{pt}}(d, T) = \Psi_{\mathfrak{D}}^{\text{unif}}(d, T) =: \Psi_{\mathfrak{D}}(d, T)$$

and

$$\frac{\gamma}{1+\gamma} \Psi_{\mathfrak{D}}(d, T) \leq \mathfrak{R}_{\mathfrak{D}}^{\text{M}}(d, T) = \mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq 1 + \frac{J_T + 1}{1-\gamma/\vartheta} \Psi_{\mathfrak{D}}(d, T). \quad (22)$$

Thus the stopped envelope is a finite-horizon minimax characterization up to a factor $O_{\gamma,\vartheta}(\log T)$.

Proof. Measurable ϵ -optimal selection gives $\mathsf{C}_{j,T}^{\text{pt}}(d) = \mathsf{C}_{j,T}^{\text{unif}}(d)$ for every shell. Apply Theorem 2.4 and then finite-horizon Bayes–minimax equality. $\square$

Why the finite-horizon form is stronger. Theorem 2.3 is convenient when one uniform bound controls every risk scale. Theorem 2.4 instead uses the cap- T cost at each scale and compares it with the cost of commitment. Both sides concern the same horizon T ; no calibration to $n \asymp \mathsf{SC}_b(Q)/b^2$ is needed. Once pointwise experiments are measurably selectable, the stopped envelope characterizes regret within an $O(\log T)$ factor. The exact upper object is the sum in (20); the logarithm appears only when that scale profile is compressed to its largest term. The polynomial-exponent identity below is therefore a coarser consequence, not the sole sense in which stopped cost is two-sided.

How the multiscale transfer works. For the upper bound, invoke a stopped experiment at the active geometric risk scale and play a Bayes action below scale 2η . After a hit at scale b_j , the risk supermartingale and Doob’s inequality bound the probability of returning above ϑb_j by γ/ϑ . Thus the expected number of visits to each scale is geometric, and summing the experiment costs gives $O_{\gamma,\vartheta}(\mathsf{SC}^{\text{unif}}/\eta)$; low-risk rounds cost at most $2T\eta$.

The lower direction contains the essential conversion. Stop an arbitrary n -round policy at $N = \tau_b \wedge n$ and write $p = \mathbb{P}_Q(\tau_b \leq n)$. If $\mathsf{SC}_b(Q) > c$, the same policy must satisfy

$$\mathbb{E}_Q R_n \geq \frac{c}{b} p, \quad \mathbb{E}_Q R_n \geq \gamma b n (1 - p). \quad (23)$$

The first inequality prices the branch that reaches the contraction set; the second prices the branch that remains at risk scale b . Minimizing their maximum over p gives (16). This is why a constant-confidence sample bound alone is insufficient: it controls neither decision cost nor a policy that contracts only on selected transcripts. Appendix A gives the full proof, including horizon truncation and the geometric-scale concatenation.

2.4 Exponent sandwich and two-sided closure

The finite-horizon envelope retains the full scale profile. Polynomial exponents give a coarser one-number summary that is convenient for dimension-dependent model classes, while preserving the distinction between statewise existence and measurable implementation. Define

$$\alpha_{\mathfrak{D}}^{\text{stop,pt}} = \sup_{m \in \mathbb{N}} \limsup_{d \rightarrow \infty} \frac{\log\{1 + \text{SC}_{\mathfrak{D}}^{\text{pt}}(d, d^m)\}}{\log d}, \quad (24)$$

$$\alpha_{\mathfrak{D}}^{\text{stop,unif}} = \sup_{m \in \mathbb{N}} \limsup_{d \rightarrow \infty} \frac{\log\{1 + \text{SC}_{\mathfrak{D}}^{\text{unif}}(d, d^m)\}}{\log d}, \quad (25)$$

$$\alpha_{\mathfrak{D}}^{\text{B}} = \sup_{M \in \mathbb{N}} \limsup_{d \rightarrow \infty} \sup_{2 \leq T \leq d^M} \frac{\log\{1 + \mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T)/\sqrt{T}\}}{\log d}. \quad (26)$$

If a frequentist minimax value $\mathfrak{R}_{\mathfrak{D}}^{\text{M}}$ is defined, let $\alpha_{\mathfrak{D}}^{\text{M}}$ be given by the same display.

Theorem 2.6 (Regret and stopped exponents). *Every sequence of repeated latent decision problems satisfying the assumptions of this section obeys*

$$\frac{\alpha_{\mathfrak{D}}^{\text{stop,pt}}}{2} \leq \alpha_{\mathfrak{D}}^{\text{B}} \leq \frac{\alpha_{\mathfrak{D}}^{\text{stop,unif}}}{2}. \quad (27)$$

Neither inequality requires convexity or a particular observation law.

Suppose, in addition, that for every finite d, B and every $\epsilon > 0$, the statewise experiments have a universally measurable choice whose ratio at every active (Q, b) is at most $\text{SC}_b(Q) + \epsilon$. Then

$$\text{SC}_{\mathfrak{D}}^{\text{pt}}(d, B) = \text{SC}_{\mathfrak{D}}^{\text{unif}}(d, B) =: \text{SC}_{\mathfrak{D}}(d, B),$$

and

$$\alpha_{\mathfrak{D}}^{\text{B}} = \frac{\alpha_{\mathfrak{D}}^{\text{stop}}}{2}. \quad (28)$$

If finite-horizon Bayes-minimax equality also holds, then

$$\alpha_{\mathfrak{D}}^{\text{M}} = \alpha_{\mathfrak{D}}^{\text{B}} = \frac{\alpha_{\mathfrak{D}}^{\text{stop}}}{2}. \quad (29)$$

Corollary 2.7 (Finite latent-model closure). *Consider a sequence of repeated latent decision problems satisfying the standing assumptions. Suppose that for every d , the latent class Ω_d and action set $\mathcal{A}_d$ are finite, although their cardinalities may grow with d . The observation space may be any standard Borel space, and the observation laws may be arbitrary Borel kernels. Then, for every finite B, T ,*

$$\text{SC}_{\mathfrak{D}}^{\text{pt}}(d, B) = \text{SC}_{\mathfrak{D}}^{\text{unif}}(d, B), \quad \mathfrak{R}_{\mathfrak{D}}^{\text{M}}(d, T) = \mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T), \quad (30)$$

where

$$\mathfrak{R}_{\mathfrak{D}}^{\text{M}}(d, T) := \inf_{\pi} \sup_{\omega \in \Omega_d} \mathbb{E}_{\omega}^{\pi} R_T.$$

Consequently,

$$\alpha_{\mathfrak{D}}^{\text{M}} = \alpha_{\mathfrak{D}}^{\text{B}} = \frac{1}{2} \alpha_{\mathfrak{D}}^{\text{stop}}. \quad (31)$$

For the *cap-T* shell profiles, $C_{j,T}^{\text{pt}}(d) = C_{j,T}^{\text{unif}}(d)$; hence the finite-horizon minimax value also satisfies (22).

The proof applies Theorem 2.3 on polynomial scale windows and calibrates $n \asymp \text{SC}_b(Q)/b^2$ on the lower side. Details are in Appendix A.

This equality does not require a separate information scale or an assumption that every query is informative. It still requires nonnegative gaps, posterior sufficiency, repetition of the same decision problem, and measurable selection.

3 Aggregate-feedback top- k bandits

We now use the conversion to obtain a finite-horizon minimax result. There are n independent arms with unit-variance Gaussian rewards. At round t , the learner selects $S_t \subset [n]$, $|S_t| = k$, and observes only

$$Y_t = \sum_{j \in S_t} X_{t,j}.$$

The reward is the same sum. Thus this is full-bandit, rather than semi-bandit, feedback. Write $\mathfrak{R}_{n,k}^*(T)$ for minimax expected regret over mean vectors in $[0, 1]^n$.

Theorem 3.1 (Dense top- k aggregate-feedback regret). *There are universal constants $c, C > 0$ such that, for $n = 2k$,*

$$c \min\{kT, k^{3/2}\sqrt{T}\} \leq \mathfrak{R}_{2k,k}^*(T) \leq C \min\{kT, k^{3/2}\sqrt{T \log(eT)}\}. \quad (32)$$

Consequently the minimax rate is $\tilde{\Theta}(\min\{kT, k^{3/2}\sqrt{T}\})$.

The upper bound is the minimum of the trivial kT bound and the polynomial-time Hadamard-design CSAR guarantee $O(k\sqrt{nT \log(eT)})$ of [Rejwan and Mansour \(2020\)](#). Their Corollary 8 displays a factor- $\sqrt{k}$ gap, up to logarithms, between this bound and the semi-bandit lower bound and leaves tighter k -dependence for further research. Theorem 3.1 closes the dense $n = 2k$, independent-Gaussian slice of that question, and Corollary 3.4 extends the conclusion throughout $n = \Theta(k)$. It does not settle the general (n, k) phase diagram when n/k diverges.

The one-context rate has close precedents, so we separate the rate claim from its broader consequences. Theorem 3.1 places the $k^{3/2}\sqrt{T}$ lower scale in the unrestricted k -subset, independent-Gaussian model of [Rejwan and Mansour \(2020\)](#); earlier lower bounds at this scale either restricted the feasible sets to one arm per group or used correlated feedback. The two-query reduction in Remark 3.2 shows that, on the paired one-context family, unrestricted and grouped queries differ by only a constant factor. The additional contribution in Section 3.1 is that the stopped-cost calculation extends to finite contextual sign classes: it gives the sharp worst-class $\log |\mathcal{F}|$ dependence and an exact factor- k local cost of aggregate rather than semi-bandit feedback on the same balanced product classes. We do not claim that the paired family supplies a new hard geometry for general (n, k) .

Table 1: Closest noncontextual benchmarks surrounding top- k aggregate feedback. Rates are in unnormalized selected-sum units. “Independent” means conditionally independent given the fixed instance.

Work	feasible actions	within-round law	result relevant here
Audibert et al. (2014)	one arm from each of k groups	independent Bernoulli coordinates	$\Omega(k\sqrt{nT})$; the same dense scale under the restriction of exactly one arm per group
Cohen et al. (2017)	grouped multitask actions	common Gaussian shock	stronger grouped lower bound from correlated coordinates; not the independent-arm model
Ito et al. (2019)	all k -subsets	strongly correlated binary coordinates	$\Omega(\sqrt{nk^3T})$ in the principal regime; the stronger scale uses correlation
Agarwal et al. (2021)	all k -subsets	possibly correlated, nonlinear aggregate reward	$\Omega(k\sqrt{knT})$ in their model; not an independent additive arm lower bound
Rejwan and Mansour (2020)	all k -subsets	independent sub-Gaussian arms	$O(k\sqrt{nT \log(eT)})$; the matching model and upper bound
This work	all k -subsets, $n \geq 2k$	independent unit-variance Gaussian arms	$\Omega(\min\{kT, k^{3/2}\sqrt{T}\})$ for every $n \geq 2k$; in the square-root regime this matches $k\sqrt{nT}$ up to logarithms when $n = \Theta(k)$. The finite-class contextual extension is stated separately below

What remains open when n/k diverges. Adding known mean-zero decoys extends the dense lower bound to every $n \geq 2k$, and therefore matches the known upper bound up to logarithms throughout the constant-density regime; see Corollary 3.4. It does not create the missing $\sqrt{n}$ dependence when $n \gg k$. For example, in a k -group, one-good-arm prior, the posterior-optimal unrestricted k -set may choose several arms from one group and omit another; its risk is governed by the k largest posterior marginals globally, not by a sum of k within-group classification risks. Thus the grouped lower bound of [Audibert et al. \(2014\)](#) does not transfer. A route to the full $k\sqrt{nT}$ lower scale would be a localized stopped adaptive-sensing inequality of the form

$$\mathbb{E}N \gtrsim \frac{pn}{\varepsilon^2}$$

for every aggregate-query experiment whose posterior risk contracts with probability p . The linear dependence on p , required because stopped cost divides by p , is exactly the unproved step. We therefore leave the diverging-density general- (n, k) problem open.

Pair the arms and draw $\theta = (\theta_1, \dots, \theta_k)$ uniformly from $\{-1, +1\}^k$. For $i \in [k]$, set

$$\mu_{i,+} = \frac{1}{2} + \varepsilon\theta_i, \quad \mu_{i,-} = \frac{1}{2} - \varepsilon\theta_i, \quad 0 < \varepsilon \leq \frac{1}{4}. \quad (33)$$

For a selected k -set S , put

$$a_i(S) = \mathbf{1}\{(i, +) \in S\} - \mathbf{1}\{(i, -) \in S\} \in \{-1, 0, 1\}.$$

Then

$$Y = \frac{k}{2} + \varepsilon \langle a(S), \theta \rangle + \xi, \quad \xi \sim N(0, k), \quad (34)$$

and the unnormalized gap is

$$\Delta_\theta^{\text{raw}}(S) = \varepsilon \{k - \langle a(S), \theta \rangle\}. \quad (35)$$

This representation covers every k -subset, including subsets that choose both arms from one pair and neither arm from another: $a(S) \in \{-1, 0, 1\}^k$ and $\|a(S)\|_2^2 \leq k$. Under any posterior, one arm in each pair has posterior mean at least $1/2$ and the other at most $1/2$; hence a Bayes-optimal k -subset chooses the posterior-preferred arm from every pair. We apply the general theory to $\Delta_\theta = \Delta_\theta^{\text{raw}}/k \in [0, 1]$.

Remark 3.2 (Dense unrestricted queries versus grouped queries). *For any $a \in \{-1, 0, 1\}^k$, choose $u, v \in \{-1, +1\}^k$ with $u + v = 2a$. These are two feasible one-arm-per-pair queries. If Y_u, Y_v are their independent aggregate observations, then*

$$\tilde{Y} = \frac{Y_u + Y_v}{2} + \zeta, \quad \zeta \sim N(0, k/2),$$

has exactly the law

$$\tilde{Y} \sim N(k/2 + \varepsilon \langle a, \theta \rangle, k)$$

of the unrestricted query, while

$$\Delta_\theta^{\text{raw}}(u) + \Delta_\theta^{\text{raw}}(v) = 2\Delta_\theta^{\text{raw}}(a).$$

Thus an unrestricted T -query policy can be simulated by a grouped $2T$ -query policy with exactly twice its regret. This explains both the closeness to [Audibert et al. \(2014\)](#) and why our novelty claim is restricted to the exact stochastic model and stopped-cost formulation.

Let $m_i(H) = \mathbb{E}[\theta_i \mid H]$. The posterior risk is therefore exactly

$$r(Q_H) = \frac{\varepsilon}{k} \sum_{i=1}^k \{1 - |m_i(H)|\}. \quad (36)$$

In particular the product prior Q_0 is active at scale $b = \varepsilon$.

Proposition 3.3 (Stopped cost of the paired family). *For every fixed $0 < \gamma < 1$, uniformly over $0 < \varepsilon \leq 1/4$,*

$$\text{SC}_\varepsilon(Q_0) = \Theta_\gamma(k). \quad (37)$$

Proof architecture. For the upper stopped bound, a fixed Rademacher design of $N = O_\gamma(k/\varepsilon^2)$ feasible sign queries has a well-conditioned Gram matrix. Least squares then has constant coordinatewise sign error. The posterior Bayes sign rule can only improve on this estimator, so Markov's inequality gives constant-probability contraction of (36). The prior expected normalized gap of every predetermined query is ε , yielding stopped cost $O_\gamma(k)$.

For the reverse bound, the stopped direct-product lemma (Lemma B.1) gives, for every experiment with success probability p ,

$$\mathbb{E}N \geq c_\gamma \frac{pk}{\varepsilon^2}. \quad (38)$$

A successful transcript has $\sum_i |m_i| \geq (1 - \gamma)k$. Flipping coordinate i changes a conditional Gaussian mean by $2\varepsilon a_i$, while $\sum_i a_i^2 \leq k$ cancels the aggregate noise variance k . Lemma 2.2 therefore turns (38) into stopped cost $\Omega_\gamma(k)$. The complete design, posterior-risk, and stopping-time arguments are in Appendix B.

What the stopped conversion contributes here. Constant-factor Hamming or Bayes-risk contraction on the paired family has sample complexity $\Theta(k/\varepsilon^2)$. That fact alone does not imply (32): a sample may be statistically informative while having very different decision loss, and a finite-horizon policy may achieve contraction only on a rare subset of transcripts. Proposition 3.3 couples information, regret, and the probability of posterior contraction for the same stopped policy. Theorem 2.3 then handles both the branch that contracts and the branch that continues to pay risk. This is precisely the step absent from a fixed-confidence sample-complexity argument. A direct horizon-wise Assouad proof, or the two-query reduction in Remark 3.2, can recover this particular lower bound. For this application, the modular contribution is therefore the following: once the model-specific stopped cost is known, the general two-branch theorem converts it into regret. Choosing $\varepsilon \asymp \sqrt{k/T}$ for $T \gtrsim k$, and a constant ε for $T \lesssim k$, gives the two regimes in Theorem 3.1.

Corollary 3.4 (Constant-density extension). *For every $n \geq 2k$,*

$$c \min\{kT, k^{3/2}\sqrt{T}\} \leq \mathfrak{R}_{n,k}^*(T) \leq C \min\{kT, k\sqrt{nT \log(eT)}\}. \quad (39)$$

Consequently, for every fixed $C_0 \geq 2$, uniformly over $2k \leq n \leq C_0 k$,

$$\mathfrak{R}_{n,k}^*(T) = \tilde{\Theta}_{C_0}(\min\{kT, k\sqrt{nT}\}).$$

For the lower bound, retain the $2k$ paired arms and add $n - 2k$ known mean-zero, unit-variance Gaussian decoys. Every paired arm has posterior mean at least $1/4$, so a Bayes action never uses a decoy. A query containing decoys has the same informative coefficient $a(S)$, still satisfies $\|a(S)\|_2^2 \leq k$, and incurs an additional nonnegative gap. Proposition 3.3 is therefore unchanged. The upper bound is again the guarantee of Rejwan and Mansour (2020). This argument does not produce the missing $\sqrt{n}$ lower dependence when n/k diverges.

3.1 Finite contextual classes and the price of aggregation

The paired calculation admits a finite-class contextual extension that also isolates the cost of losing individual attribution. Let $\mathcal{X}$ be a finite context space, let $X_t \stackrel{\text{iid}}{\sim} \nu$ independently of the latent model, and reveal X_t before the action. Fix a known $0 < \varepsilon \leq 1/4$ and a finite class

$$\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \{-1, +1\}^k\}.$$

The symmetric baseline $1/2$, the margin ε , the context law ν , and the class $\mathcal{F}$ are known to the learner. For $f^* \in \mathcal{F}$, context x , pair i , and $s \in \{-1, +1\}$, arm (i, s) has mean

$$\mu_{x,i,s} = \frac{1}{2} + s\varepsilon f_i^*(x) \quad (40)$$

and unit-variance Gaussian noise, mutually independent across arms and rounds conditional on f^* . The learner chooses any k -subset and initially observes only its sum. The comparator chooses $\{(i, f_i^*(x)) : i \in [k]\}$ at context x .

Formally, an action in the repeated latent problem is a contextual decision rule $x \mapsto S(x)$. A randomized contextual policy can be drawn before X_t and evaluated only at the realized context, so this macro-action representation has the same value as the usual protocol in which X_t is observed before S_t is selected. With

$$a_i(S) = \mathbf{1}\{(i, +1) \in S\} - \mathbf{1}\{(i, -1) \in S\},$$

the observation and raw gap are

$$Y_t - \frac{k}{2} = \varepsilon \langle a(S_t), f^*(X_t) \rangle + \xi_t, \quad \xi_t \sim N(0, k), \quad (41)$$

$$\Delta_{f^*}^{\text{raw}}(X_t, S_t) = \varepsilon \{k - \langle a(S_t), f^*(X_t) \rangle\}. \quad (42)$$

We again divide the gap by k when applying the general theory. For a posterior Q on $\mathcal{F}$, put $m_{Q,i}(x) = \mathbb{E}_Q f_i(x)$. Its Bayes risk is

$$r(Q) = \frac{\varepsilon}{k} \mathbb{E}_{X \sim \nu} \sum_{i=1}^k \{1 - |m_{Q,i}(X)|\}. \quad (43)$$

Write $\mathfrak{R}_{\mathcal{F}, \varepsilon}^{\text{agg}}(T)$ for the frequentist minimax regret in raw selected-sum units for this known $(\nu, \mathcal{F}, \varepsilon)$ problem.

Proposition 3.5 (Finite-class contextual upper certificate). *For every fixed $0 < \gamma < 1$, there is $C_\gamma < \infty$ such that, for every finite $\mathcal{F}$, every prior Q on $\mathcal{F}$, and every active scale b ,*

$$\text{SC}_b^{\text{agg}}(Q) \leq C_\gamma \log(e|\mathcal{F}|). \quad (44)$$

Consequently, for the raw selected-sum regret,

$$\mathfrak{R}_{\mathcal{F}, \varepsilon}^{\text{agg}}(T) \leq C_\gamma k \min \left\{ \varepsilon T, 1 + \sqrt{T \log(e|\mathcal{F}|)} \right\}. \quad (45)$$

The experiment behind (44) is deliberately simple. At every context it selects one arm per pair using independent Rademacher signs. For

$$h_\nu(f, g) = \mathbb{E}_{X \sim \nu} d_H(f(X), g(X)),$$

the Hellinger affinity of one aggregate observation under f and g is at most

$$\exp \left\{ -c \frac{\varepsilon^2}{k} h_\nu(f, g) \right\}.$$

The maximum-likelihood estimate from N such queries therefore satisfies

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f h_\nu(f, \hat{f}) \leq C \min \left\{ k, \frac{k \log(e|\mathcal{F}|)}{N \varepsilon^2} \right\}.$$

Posterior Bayes signs can only improve on this estimator. Taking $N \asymp_\gamma \log(e|\mathcal{F}|)/(\varepsilon b)$ makes (43) contract with constant probability. Every random grouped query has expected normalized gap exactly ε , which proves the stopped-cost bound. The full Hellinger and stopping argument is in Appendix C.

The upper bound is sharp for balanced classes, but cardinality alone does not lower-bound every individual problem: two functions may differ only on a ν -null or very rare context, and many coordinates may move coherently. The following product class supplies the matching geometry. Let $\mathcal{X} = [q]$, let ν be uniform, and define

$$\mathcal{F}_{q,k} = \{f_\theta : f_\theta(x) = \theta_x, \theta \in \{-1, +1\}^{q \times k}\}. \quad (46)$$

Write

$$d = qk = \log_2 |\mathcal{F}_{q,k}|,$$

and let Q_0 be uniform on this class. For this product class, use $\mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T)$ and $\mathfrak{R}_{q,k,\varepsilon}^{\text{semi}}(T)$ for the corresponding raw minimax values under aggregate and semi-bandit feedback. The superscripts on both $\mathfrak{R}$ and SC distinguish these two observation kernels; the action set and regret gaps are identical.

Proposition 3.6 (Exact contextual cost and feedback price). *Uniformly over $0 < \varepsilon \leq 1/4$,*

$$\text{SC}_\varepsilon^{\text{agg}}(Q_0) = \Theta_\gamma(qk) = \Theta_\gamma(\log |\mathcal{F}_{q,k}|). \quad (47)$$

If the learner instead observes the individual rewards of all selected arms, with the same action set, models, and gaps, then

$$\text{SC}_\varepsilon^{\text{semi}}(Q_0) = \Theta_\gamma(q) = \Theta_\gamma\left(\frac{\log |\mathcal{F}_{q,k}|}{k}\right). \quad (48)$$

Thus removing attribution multiplies the exact local certificate by k . More explicitly, the two-branch theorem and elementary explore-commit upper bounds give, up to logarithmic factors,

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T) = \tilde{\Theta}\left(k \min\left\{\varepsilon T, \frac{qk}{\varepsilon}\right\}\right), \quad (49)$$

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{semi}}(T) = \Theta\left(k \min\left\{\varepsilon T, \frac{q}{\varepsilon}\right\}\right). \quad (50)$$

When $T \gtrsim qk$, at the aggregate hard scale $\varepsilon \asymp \sqrt{qk/T}$, the aggregate and semi-bandit values are respectively

$$\Theta(k\sqrt{qkT}) \quad \text{and} \quad \Theta(\sqrt{qkT}),$$

so they differ by an exact factor of order k . Maximizing each feedback model over its own signal scale gives

$$\sup_{0 < \varepsilon \leq 1/4} \mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T) = \Theta\left(k \min\{T, \sqrt{qkT}\}\right), \quad (51)$$

$$\sup_{0 < \varepsilon \leq 1/4} \mathfrak{R}_{q,k,\varepsilon}^{\text{semi}}(T) = \Theta\left(k \min\{T, \sqrt{qT}\}\right). \quad (52)$$

Hence these worst-known-scale envelopes differ by $\Theta(\sqrt{k})$ once both are in their square-root regimes; the learner is told ε in each fixed-scale problem before the supremum is taken. These comparisons use identical unrestricted k -subset actions and independent arm noises; only the observation changes.

For a cardinality-only summary, let $\mathfrak{V}_{k,M}^{\text{agg}}(T)$ be the supremum of the raw minimax value over the finite contextual paired problems above with $|\mathcal{F}| \leq M$, where $\nu, \mathcal{F}$, and ε are known to the learner. For $M \geq 2^k$,

$$ck \min\{T, \sqrt{T \log M}\} \leq \mathfrak{V}_{k,M}^{\text{agg}}(T) \leq Ck \min\{T, 1 + \sqrt{T \log(eM)}\}. \quad (53)$$

The lower bound chooses the largest balanced product class of size at most M and tunes ε to the horizon; the upper bound is Proposition 3.5. Thus $\log |\mathcal{F}|$ is the exact worst finite-class dependence, while Proposition 3.6 explains when an individual class attains it.

Contextual aggregate feedback itself has precedents. Zierahn et al. (2023) study adversarial bilinear contextual combinatorial bandits under both aggregate full-bandit and semi-bandit feedback. The finite-policy and general-function work of Erez and Koren (2025) and Qin and Zhang (2026) centers on semi-bandit feedback; Erez and Koren (2025) identify the general full-bandit list-learning extension as open, while Qin and Zhang (2026) also analyze generic aggregate-feedback reductions. Our result is neither the first contextual aggregate algorithm nor a solution of that broader open problem. Its contribution is the exact regret-priced contraction law, matching lower bounds, and aggregate-versus-semi separation for the stochastic paired, known-margin Gaussian class. The finite-class upper uses maximum likelihood over $\mathcal{F}$ and makes no oracle-efficiency claim.

4 Linear and combinatorial preference feedback

We next calculate stopped cost for a finite linear preference class. The result permits continuous linear queries, contains the deterministic preference chain as a special case, and also covers the full dense top- d action set rather than only one-choice-per-pair queries.

For $d \geq 1$, put

$$\mathcal{H}_d = \left\{ d^{-1/2} s : s \in \{-1, +1\}^d \right\},$$

and let $\mathcal{X}$ be any compact action set satisfying

$$\mathcal{H}_d \subseteq \mathcal{X} \subseteq B_2^d(1). \quad (54)$$

For $s \in \{-1, +1\}^d$, the latent utility parameter is

$$\theta_s = \frac{b}{\sqrt{d}} s, \quad 0 < b \leq \frac{1}{2}.$$

Thus $\max_{x \in \mathcal{X}} \langle \theta_s, x \rangle = b$. One query chooses $x, x' \in \mathcal{X}$ and observes

$$\mathbb{P}_s(Y = 1 \mid x, x') = F(\beta \langle \theta_s, x - x' \rangle), \quad F \in \{\Phi, \sigma\}, \quad \beta > 0, \quad (55)$$

where Φ is the standard Gaussian cdf and $\sigma(u) = (1 + e^{-u})^{-1}$. The pair-average utility gap is

$$\Delta_s(x, x') = b - \frac{1}{2} \langle \theta_s, x + x' \rangle. \quad (56)$$

It lies in $[0, 2b]$, and it charges the utility loss of both queried actions through their average. Thus the total regret of the two actions used by one comparison is $2\Delta_s(x, x')$.

Let Q_0 be uniform on the 2^d parameters. If $m_i(h) = \mathbb{E}[s_i \mid h]$, $\bar{\theta}_h = \mathbb{E}[\theta_s \mid h]$, and $h_{\mathcal{X}}(v) = \sup_{x \in \mathcal{X}} \langle v, x \rangle$, then

$$r_{\mathcal{X}}(Q_h) = b - h_{\mathcal{X}}(\bar{\theta}_h), \quad \bar{\theta}_h = \frac{b}{\sqrt{d}} m(h), \quad r_{\mathcal{X}}(Q_0) = b. \quad (57)$$

Proposition 4.1 (Stopped cost for linear preference classes). *Fix $0 < \gamma < 1$, $F \in \{\Phi, \sigma\}$, and $\beta > 0$. Uniformly over every action set satisfying (54),*

$$\text{SC}_b^{\text{pref}}(Q_0) = \Theta_{\gamma, F} \left(b^2 d + \frac{d^2}{\beta^2} \right) = \Theta_{\gamma, F} (b^2 D_{d, b, \beta}), \quad D_{d, b, \beta} = d + \frac{d^2}{\beta^2 b^2}. \quad (58)$$

With $d, b, \beta, F, \mathcal{X}$ known to the learner, define

$$\mathfrak{R}_{d, b, \beta, \mathcal{X}}^{\text{pref}}(T) = \inf_{\pi} \sup_{s \in \{-1, +1\}^d} \mathbb{E}_s^{\pi} \sum_{t=1}^T \Delta_s(X_t, X'_t).$$

For every $T \geq 2$,

$$c_{\gamma, F} b \min\{T, D_{d, b, \beta}\} \leq \mathfrak{R}_{d, b, \beta, \mathcal{X}}^{\text{pref}}(T) \leq C_F b \min\{T, D_{d, b, \beta} \log(eT)\}. \quad (59)$$

Corollary 4.2 (Unknown-scale minimax regret). *Fix $b_0 \in (0, 1/2]$. Let the learner know d, b_0, β , the link, and the action set, but not the realized $b \in (0, b_0]$. Define*

$$\mathfrak{R}_{d, \leq b_0, \beta, \mathcal{X}}^{\text{pref}}(T) = \inf_{\pi \text{ independent of } b} \sup_{\substack{0 < b \leq b_0 \\ s \in \{-1, +1\}^d}} \mathbb{E}_{s, b}^{\pi} \sum_{t=1}^T \Delta_{s, b}(X_t, X'_t).$$

Then

$$\begin{aligned} c_{\gamma,F} \min \left\{ b_0 T, b_0 d + \frac{d}{\beta} \sqrt{T} \right\} &\leq \mathfrak{R}_{d,\leq b_0,\beta,\mathcal{X}}^{\text{pref}}(T) \\ &\leq C_F \min \left\{ b_0 T, b_0 d \log(eT) + \frac{d}{\beta} \sqrt{T \log(eT)} \right\}. \end{aligned} \quad (60)$$

Thus, up to logarithms,

$$\mathfrak{R}_{d,\leq b_0,\beta,\mathcal{X}}^{\text{pref}}(T) = \tilde{\Theta} \left(\min \left\{ b_0 T, b_0 d + \frac{d}{\beta} \sqrt{T} \right\} \right).$$

For constant b_0 and $\beta = \Theta(1)$, this is $\tilde{\Theta}(\min\{T, d\sqrt{T}\})$.

Proof. For $c = d^2/\beta^2$, elementary optimization gives

$$\sup_{0 < b \leq b_0} b \min \left\{ T, d + \frac{c}{b^2} \right\} \asymp \min \left\{ b_0 T, b_0 d + \sqrt{cT} \right\}.$$

For the upper bound, the left side is at most $b_0 T$, while

$$b \min \left\{ T, d + \frac{c}{b^2} \right\} \leq b_0 d + \min \left\{ bT, \frac{c}{b} \right\} \leq b_0 d + \sqrt{cT}.$$

For the reverse bound, choose $b = b_0$, and, when it lies below b_0 , also choose

$$b = \sqrt{\frac{c}{T-d}}$$

for $T > d$. Splitting into $T \leq 2d$, $T > 2d$ with this second choice feasible, and $T > 2d$ without it proves the display up to an absolute factor. The fixed- b lower bound in (59), followed by the supremum over b , proves the first inequality.

For the upper bound, use the following one policy for every b . At exploratory round n , draw a uniform Rademacher vector ξ_n , compare the antipodal vertices $\xi_n/\sqrt{d}$ and $-\xi_n/\sqrt{d}$, and put

$$Z_{n,i} = \xi_{n,i}(2Y_n - 1).$$

Let

$$r_n = \sqrt{\frac{2 \log(4dT^3)}{n}}.$$

Stop exploring at the first $n \leq T$ such that $\min_i |n^{-1} \sum_{t \leq n} Z_{t,i}| > 2r_n$, and thereafter play the vertex given by the empirical coordinate signs against itself. The correlation bound proved below gives, under every s, b ,

$$|\mathbb{E}_{s,b} Z_{n,i}| \geq c_F \min \left\{ \frac{\beta b}{d}, \frac{1}{\sqrt{d}} \right\}, \quad \text{sign}(\mathbb{E}_{s,b} Z_{n,i}) = s_i.$$

A union of Hoeffding bounds over i and $n \leq T$ has probability at least $1 - T^{-2}$. On this event the policy stops with the correct sign vector after at most

$$C_F \left(d + \frac{d^2}{\beta^2 b^2} \right) \log(eT)$$

queries, unless this exceeds T . Each exploratory antipodal pair has gap exactly b , exploitation has zero gap on the event, and the pair gap is always at most $2b$. Hence the same policy, without knowing b , satisfies

$$\mathbb{E}_{s,b} R_T \leq C_F b \min\{T, D_{d,b,\beta} \log(eT)\}.$$

Optimize this expression as in the first display, replacing d, c by $d \log(eT), c \log(eT)$. $\square$

Why the action-set uniformity matters. The lower bound allows every comparison in the Euclidean ball. A contraction transcript must satisfy

$$h_{\mathcal{X}}(\bar{\theta}_h) \geq (1 - \gamma)b.$$

Because $\mathcal{X} \subseteq B_2^d(1)$, this forces $\|m(h)\|_2^2 \geq (1 - \gamma)^2 d$. The binary transcript then requires $\Omega_\gamma(pd)$ labels on a contraction event of probability p . A coordinate-flip argument gives the second, sensitivity-dependent requirement

$$\mathbb{E}N \gtrsim_{\gamma,F} p \frac{d^2}{\beta^2 b^2}.$$

Indeed, flipping s_i changes the link argument by $2\beta b(x_i - x'_i)/\sqrt{d}$, while $\|x - x'\|_2 \leq 2$. Before contraction, each query has posterior expected gap at least γb , which gives the lower half of (58).

For the upper bound, only vertices of $\mathcal{H}_d$ are needed. Compare antipodal random Rademacher vertices. If $\alpha = \beta b/d$, their signed labels have coordinate correlation at least

$$c_F \min\{\alpha, d^{-1/2}\}.$$

Thus $O_{\gamma,F}(d + \alpha^{-2}) = O_{\gamma,F}(D_{d,b,\beta})$ labels suffice. The first term is the one-bit floor after the link saturates, not a global link-condition penalty. Every comparison has gap exactly b , and the same design supports the unknown- b stopping rule in Corollary 4.2. This proves (58); an explore-commit version gives (59).

Three specializations. *Linear dueling.* Taking $\mathcal{X} = B_2^d(1)$ gives a continuous-action linear preference problem with a finite hypercube parameter class. The two terms in (58) have distinct meanings: $b^2 d$ is the one-bit floor, whereas d^2/β^2 is the weak-signal experimental-design cost. In particular, for fixed β and $\beta b \leq \sqrt{d}$,

$$\mathfrak{R}_{d,b,\beta,B_2^d}^{\text{pref}}(T) = \tilde{\Theta}\left(\min\left\{bT, \frac{d^2}{\beta^2 b}\right\}\right).$$

Corollary 4.2 recovers the familiar $\tilde{\Theta}(d\sqrt{T}/\beta)$ linear-preference scale in the weak-signal, nontrivial-horizon regime; general linear trajectory-preference regret under the same pair-average convention is studied by Saha et al. (2023).

Deterministic trajectory preferences. Taking $\mathcal{X} = \mathcal{H}_d$ gives the policy class of a deterministic chain with d binary decisions. For the notation

$$V_s(x) = \frac{H}{2} + \varepsilon \langle x, s \rangle, \quad x \in \{-1, +1\}^d, \quad H \geq d, \quad 0 < \varepsilon \leq \frac{H}{2d},$$

the rescaling $x \mapsto x/\sqrt{d}$ and

$$b = \frac{\varepsilon d}{H}$$

turns (55) and (56) into normalized trajectory preference and pair-average value regret. One label executes two trajectories, and their total normalized value regret is twice the displayed pair-average quantity. Proposition 4.1 therefore yields

$$\text{SC}_b^{\text{pref}}(Q_0) = \Theta_{\gamma,F}\left(\frac{\varepsilon^2 d^3}{H^2} + \frac{d^2}{\beta^2}\right), \quad D = d + \frac{H^2}{\beta^2 \varepsilon^2}.$$

Full dense top- d selection. Index $2d$ items by $(i, +)$ and $(i, -)$. For an arbitrary d -subset A , define its imbalance vector

$$x(A) = \frac{1}{\sqrt{d}} (\mathbf{1}\{(i, +) \in A\} - \mathbf{1}\{(i, -) \in A\})_{i=1}^d.$$

Then the full hypersimplex image

$$\mathcal{X}_{\text{top}} = \{x(A) : A \subseteq [d] \times \{-, +\}, |A| = d\}$$

satisfies $\mathcal{H}_d \subseteq \mathcal{X}_{\text{top}} \subseteq B_2^d(1)$. Thus Proposition 4.1 applies to *all* d -subsets, including those taking both items from one pair and neither from another.

Concretely, let

$$U_s(A) = \frac{d}{2} + b \sum_{i=1}^d s_i (\mathbf{1}\{(i, +) \in A\} - \mathbf{1}\{(i, -) \in A\}),$$

and use the normalized preference model

$$\mathbb{P}_s(Y = 1 \mid A, A') = F\left(\beta \frac{U_s(A) - U_s(A')}{d}\right).$$

The theorem gives the exact stopped-cost order

$$\Theta_{\gamma, F}\left(b^2 d + \frac{d^2}{\beta^2}\right)$$

and (59) for normalized pair-average regret. Raw pair-average sum-utility regret is d times that value, while the total regret of the two selected subsets is $2d$ times that value. If a preference is generated by comparing independent scores $G_s(A) \sim N(U_s(A), d)$, then $F = \Phi$ and $\beta = \sqrt{d/2}$. For $b \leq 1/4$,

$$\text{SC}_b^{\text{pref}}(Q_0) = \Theta_{\gamma}(d), \quad \mathfrak{R}_{\text{raw, pair}}^{\text{pref}}(T) = \tilde{\Theta}\left(\min\left\{dbT, \frac{d^2}{b}\right\}\right).$$

Corollary 4.2 supplies one policy that does not know b , and gives the class-level raw pair-average minimax rate

$$\tilde{\Theta}\left(\min\{dT, d^{3/2}\sqrt{T}\}\right)$$

when b_0 is a fixed constant. Total raw regret over the $2T$ selected subsets is exactly twice this quantity. For a Bradley–Terry evaluator with fixed slope, the d^2/β^2 term dominates precisely when $\beta b = O(\sqrt{d})$, not merely when $\beta b = O(d)$.

Scope. Proposition 4.1 is a finite linear and combinatorial preference theorem, not a general tabular-RL theorem. Its distinct output is the hit-probability-sensitive stopped-cost calculation, including the one-bit floor, the evaluator-sensitivity term, and a lower bound that remains valid even when every Euclidean-ball comparison is allowed. The resulting cumulative-regret order is consistent with general linear-dueling theory. The trajectory specialization has known deterministic dynamics and is a structured linear/combinatorial dueling problem, not a new general PbRL rate (Xu et al., 2020; Saha et al., 2023; Wang et al., 2023). Unknown transitions, reachability-dependent policy features, changing evaluators, and off-policy generalization require additional arguments.

5 Gaussian feedback on compact action spaces

We next identify a broad class in which pointwise stopped experiments can be selected measurably as functions of the posterior and worst-prior Bayes regret coincides with frequentist minimax regret. Let $\mathcal{C}_d$ be a collection of nonempty compact metric action spaces. For $C \in \mathcal{C}_d$, let $\mathfrak{F}_d(C)$ be a nonempty Borel subset of $\mathcal{C}(C, [0, 1])$, with the uniform topology. If f is fixed, querying $x \in C$ reveals

$$O = f(x) + \xi, \quad \xi \sim N(0, 1),$$

and incurs the gap $f(x) - \min_C f$. Neither C nor f is assumed to be convex in this section. Write

$$\mathfrak{R}_{\mathfrak{F}}^M(d, T) = \sup_{C \in \mathcal{C}_d} \inf_{\pi} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f^{\pi} R_T, \quad (61)$$

$$\mathfrak{R}_{\mathfrak{F}}^B(d, T) = \sup_{C \in \mathcal{C}_d} \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi} \mathbb{E}_Q^{\pi} R_T. \quad (62)$$

Both infima are over Borel randomized nonanticipating policies. The values $\text{SC}_{\mathfrak{F}, C}^{\text{pt}}$ and $\text{SC}_{\mathfrak{F}, C}^{\text{unif}}$ are defined by (9)–(10); an outer supremum over $C \in \mathcal{C}_d$ gives $\text{SC}_{\mathfrak{F}}^{\text{pt}}$ and $\text{SC}_{\mathfrak{F}}^{\text{unif}}$.

Theorem 5.1 (Gaussian statewise selection and minimax equality). *For every finite d, T, B ,*

$$\mathfrak{R}_{\mathfrak{F}}^M(d, T) = \mathfrak{R}_{\mathfrak{F}}^B(d, T), \quad (63)$$

$$\text{SC}_{\mathfrak{F}, C}^{\text{pt}}(d, B) = \text{SC}_{\mathfrak{F}, C}^{\text{unif}}(d, B). \quad (64)$$

The supremum in (63) may be restricted to finitely supported priors. For every $\epsilon > 0$, the second equality is implemented by a universally measurable state-dependent experiment whose finite deterministic cap may depend on the state. For every finite cap L , the analogous capped pointwise and uniform stopped profiles also coincide.

The first equality is a finite-horizon minimax theorem, not an asymptotic interchange. The second identifies the statewise and uniform stopped values. Convexity is not needed. Compactness and continuity ensure that Bayes actions exist; lower-semianalytic dynamic programming supplies universally measurable near-minimizing experiments; and the strictly positive Gaussian density makes the posterior update Borel. Write their common worst-space value as $\text{SC}_{\mathfrak{F}}(d, B)$, and use the exponents from (24)–(26) with $\mathfrak{D} = \mathfrak{F}$. The capped-selection part of the theorem also gives $\text{C}_{j, T}^{\text{pt}}(d) = \text{C}_{j, T}^{\text{unif}}(d)$ for every shell. Therefore the Gaussian minimax value satisfies the finite-horizon characterization (22), not only the exponent identity below.

Corollary 5.2 (Regret and stopped exponents under Gaussian feedback). *For every sequence of the Gaussian loss models above,*

$$\alpha_{\mathfrak{F}}^M = \alpha_{\mathfrak{F}}^B = \frac{\alpha_{\mathfrak{F}}^{\text{stop}}}{2}. \quad (65)$$

No information-scale matching or nondegeneracy assumption is required.

Proof. Apply Theorem 2.6 and (63)–(64). □

5.1 The full convex class

We now return to stochastic BCO: $C \subset \mathbb{R}^d$ is a compact convex body and $\mathfrak{F}_d(C)$ is the full class of continuous $[0, 1]$ -valued convex losses. Convexity first becomes essential here, through the model class and the BCO bounds used below.

Corollary 5.3 (Full-class exponents). *For the full class of continuous $[0, 1]$ -valued convex losses,*

$$\frac{5}{2} \leq \alpha_{\text{full}}^{\text{stop}} \leq 3, \quad \frac{5}{4} \leq \alpha_{\text{full}}^{\text{reg}} \leq \frac{3}{2}. \quad (66)$$

Determining either exponent determines the other.

The upper endpoint follows from the current general BCO upper bounds (Lattimore and György, 2023; Fokkema et al., 2024); the lower endpoint follows from Rajaraman (2026). Appendix H records the noise-preserving rescaling needed to place the latter hard family in the present model.

6 Observation-adaptive experiments: the dimple family

Let $C = B_2^d(1)$. Fix

$$d^{-1} \leq \varepsilon \leq \varepsilon_0, \quad s = \sqrt{1 + 2\varepsilon},$$

where $\varepsilon_0 > 0$ is a sufficiently small constant. For $\theta \in S_2^{d-1}(1)$, define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}(s - \|\theta - x\|)_+^2. \quad (67)$$

These functions are continuous, convex, and $[0, 1]$ -valued after reducing ε_0 if necessary (Bakhtiari et al., 2025, Lemma 10). Their unique minimizers are θ , with $F_\theta(\theta) = 0$. At the origin,

$$b_0 := F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1 + 2\varepsilon} + 1} \asymp \varepsilon. \quad (68)$$

Bakhtiari et al. (2025) use this family to exhibit an obstruction to standard one-step information-ratio analysis. We show that the same geometry admits a cheaper multistep experiment. The lower bound below uses mutual information to show that an unoriented ray reveals very little; the upper construction uses early observations to orient later tangent probes.

Use the rotation-invariant prior $\Theta \sim \text{Unif}(S_2^{d-1}(1))$. Since $\nabla F_\theta(0) = -b_0\theta$, convexity gives

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x \rangle\}.$$

The origin is therefore the prior Bayes action and $r(Q_0) = b_0$. Write

$$\tau_\gamma = \inf\{t \geq 1 : r(Q_t) \leq \gamma b_0\}. \quad (69)$$

Definition 6.1 (Fresh directions). *A policy uses fresh directions if, at round t , it first draws $B_t \sim \text{Unif}(S_2^{d-1}(1))$, independently of (Θ, H_{t-1}) , then chooses an arbitrary $\sigma(H_{t-1}, B_t)$ -measurable radius $R_t \in [0, 1]$ and queries $X_t = R_t B_t$.*

The radius may use the complete past. Only the direction must be fresh. Every ray starts at the origin. The comparison excludes shifted or paired probes, anisotropic directions, and reuse of a learned direction.

Definition 6.2 (Block-fresh subspaces). *Fix $1 \leq m \leq d$ and $\ell \geq 1$. A policy is block-fresh of type (m, ℓ) if it proceeds in nonempty blocks. At the start of each block it draws*

$$U \sim \text{Haar}\{\text{Gr}(m, d)\}$$

conditionally independently of (Θ, H_{pre}) , and then makes an arbitrary observation-adaptive sequence of at most ℓ queries in $U \cap B_2^d(1)$. Its within-block directions, radii, randomization, and stopping decision may use the complete pre-block history, the realized subspace, and all observations already obtained in that block. A new independent subspace is drawn for the next block.

Unlike a nonadaptive batch, a block-fresh policy may adapt and reuse directions within each block. Unlike the unrestricted construction below, it cannot orient a later subspace using what earlier blocks have learned.

Theorem 6.3 (Fresh-subspace information bound). *For $d \geq 2$, there is a universal $C < \infty$ such that, for every posterior ν of Θ , every fresh $B \sim \text{Unif}(S_2^{d-1}(1))$, every measurable $R = R(B) \in [0, 1]$, and independent $\xi \sim N(0, 1)$,*

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq C(\varepsilon^4 + d^{-4}). \quad (70)$$

Under the spherical prior, every nonadaptive batch $X_1, \dots, X_n$, possibly randomized but independent of Θ and fixed before its observations, satisfies

$$I(\Theta; O_{1:n} \mid X_{1:n}) \leq Cn(\varepsilon^4 + d^{-4}). \quad (71)$$

For any initial prior, every stopped fresh-direction experiment with duration N and $\mathbb{E}N < \infty$ satisfies

$$I(\Theta; H_N) \leq C(\varepsilon^4 + d^{-4}) \mathbb{E}N. \quad (72)$$

For $1 \leq m \leq d$, put

$$A_{d,m,\varepsilon} = \varepsilon^4 + \frac{m(m+2)(m+4)(m+6)}{d(d+2)(d+4)(d+6)}. \quad (73)$$

Every stopped block-fresh experiment of type (m, ℓ) satisfies

$$I(\Theta; H_N) \leq C\ell A_{d,m,\varepsilon} \mathbb{E}N. \quad (74)$$

This bound is uniform over the initial prior.

Conversely, under the spherical initial prior, if $\mathbf{M} \in \sigma(H_N)$ is an event on which the terminal posterior Bayes risk is at most γb_0 , then

$$I(\Theta; H_N) \geq \frac{d-1}{2} (1-\gamma)^2 \mathbb{P}(\mathbf{M}). \quad (75)$$

Consequently, every stopped fresh-direction experiment under the spherical prior with $\mathbb{P}(\mathbf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathbf{M})} \geq c_\gamma \frac{d}{\varepsilon^4 + d^{-4}}. \quad (76)$$

Every block-fresh experiment of type (m, ℓ) with $\mathbb{P}(\mathbf{M}) > 0$ similarly satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathbf{M})} \geq c_\gamma \frac{d}{\ell A_{d,m,\varepsilon}}. \quad (77)$$

Under the spherical prior, (76) also holds with $N = n$ for an arbitrary nonadaptive batch of size n . Thus, under that prior, constant-probability contraction requires batch size, or expected fresh-direction duration, $\Omega_\gamma(d/(\varepsilon^4 + d^{-4}))$.

Remark 6.4 (Conditionally diffuse subspaces). *The Haar assumption in the block statement can be weakened. Conditional on each pre-block history, suppose U is independent of Θ and its possibly history-dependent law satisfies*

$$\sup_{\|v\|_2=1} \mathbb{E}[\|P_U v\|_2^8 \mid H_{\text{pre}}] \leq \kappa \left(\frac{m}{d}\right)^4.$$

Then (74) holds with $A_{d,m,\varepsilon}$ replaced by $\varepsilon^4 + \kappa(m/d)^4$, and the corresponding query, stopped-cost, and regret lower bounds follow unchanged. Thus conditional angular diffuseness, rather than literal Haar sampling, is the operative restriction.

Fix a sufficiently large $C_L = C_L(\gamma)$, and for $0 < a \leq b_0$ put

$$L_{d,a} = 1 + \log \left\{ \frac{C_L d (1 + \log(d/a))}{a} \right\}. \quad (78)$$

Theorem 6.5 (Observation-adaptive directional experiment). *There are constants $\gamma_0, c_0 > 0$, with $\gamma_0 + c_0 < \gamma$, and $C_\gamma < \infty$ such that, for every sufficiently large d , every $d^{-1} \leq \varepsilon \leq \varepsilon_0$, and every $0 < a \leq b_0$, an observation-adaptive experiment returns $\hat{x} \in B_2^d(1)$ with*

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{P}_\theta \{F_\theta(\hat{x}) > \gamma_0 a\} \leq c_0 a, \quad (79)$$

$$N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a^2}, \quad (80)$$

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{E}_\theta R_N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a}. \quad (81)$$

The query cap is deterministic. The conditional distributions and subspaces of the directions in the last two stages depend measurably on earlier observations. Under every prior on Θ ,

$$\mathbb{P} \{ \mathbb{E}[F_\Theta(\hat{x}) \mid H_N] \leq \gamma a \} \geq 1 - \frac{\gamma_0 + c_0}{\gamma} > 0. \quad (82)$$

The construction has three stages. The table suppresses logarithmic factors.

stage	progress	(queries, regret)
orthogonal scan	obtain v_0 with $\langle \theta, v_0 \rangle = \tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$	$(d^2/\varepsilon^2, d^2/\varepsilon)$
tangent amplification	choose tangent directions from the current estimate and enter a fixed angular cone	$(d^2/\varepsilon^2, d^2/\varepsilon)$
angular contraction	invert boundary observations and repeat $q \mapsto q/2$ until $q = O(a)$	$(d^2/a^2, d^2/a)$

Only the last two stages orient later probes using earlier observations. Their detailed analysis is in Appendix G.

For the next statement, $\text{SC}_b^{[1]}$ restricts (5) to one query, and $\text{SC}_b^{\text{fresh}}$ restricts it to fresh-direction policies. Write $\text{SC}_b^{\text{bf}(m,\ell)}$ for the restriction to block-fresh policies of type (m, ℓ) .

Corollary 6.6 (Polynomial separations). *For all sufficiently large d , at $\varepsilon = d^{-1}$ and $b = b_0 \asymp d^{-1}$, the initial spherical state satisfies*

$$\text{SC}_b^{[1]} = \Omega_\gamma(d^3), \quad \text{SC}_b^{\text{fresh}} = \Omega_\gamma(d^3), \quad \text{SC}_b = \tilde{O}_\gamma(d^2). \quad (83)$$

At the same scale and under the spherical prior, a nonadaptive batch with constant contraction probability has size $\Omega_\gamma(d^5)$, and every stopped fresh-direction experiment with $\mathbb{P}(\mathbf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathbf{M})} \geq c_\gamma d^5.$$

In particular, constant success probability requires $\mathbb{E}N = \Omega_\gamma(d^5)$. The observation-adaptive construction contracts risk with constant probability using $\tilde{O}_\gamma(d^4)$ queries.

Let

$$L_{d,T} = 1 + \log \{ C_L d \max\{T, d\} [1 + \log\{d \max\{T, d\}\}] \}.$$

Under the spherical prior, every fresh-direction policy satisfies

$$\mathbb{E}R_T \geq c_\gamma \min \left\{ \frac{T}{d}, d^4 \right\}, \quad (84)$$

while an unrestricted policy satisfies, uniformly in θ ,

$$\mathbb{E}_\theta R_T \leq C_\gamma L_{d,T} \min \left\{ \frac{T}{d}, d\sqrt{T} \right\}. \quad (85)$$

At $T = d^5$, these bounds are $\Omega(d^4)$ and $\tilde{O}(d^{7/2})$, respectively.

Corollary 6.7 (Growing block-fresh separation). *Let $1 \leq m \leq \varepsilon_0 d$, $\ell \geq 1$, and set $\varepsilon = m/d$, so $b = b_0 \asymp m/d$. Then*

$$\text{SC}_b^{\text{bf}(m,\ell)} \geq c_\gamma \frac{d^3}{\ell m^2}, \quad \text{SC}_b \leq \tilde{O}_\gamma(d^2), \quad (86)$$

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathbf{M})} \geq c_\gamma \frac{d^5}{\ell m^4} \quad \text{for every successful block-fresh experiment,} \quad (87)$$

whereas the unrestricted experiment contracts with constant probability using $\tilde{O}_\gamma(d^4/m^2)$ queries.

Under the spherical prior, every block-fresh policy of type (m, ℓ) satisfies

$$\mathbb{E}R_T \geq c_\gamma \min \left\{ \frac{mT}{d}, \frac{d^4}{\ell m^3} \right\}. \quad (88)$$

If, for some fixed $\delta > 0$, $\ell m^2 \leq d^{1-\delta}$ and $T_\star \asymp d^5/(\ell m^4)$, then the restricted lower bound is $\Omega_\gamma(d^4/(\ell m^3))$, while the unrestricted construction has regret

$$\tilde{O}_\gamma \left(\frac{d^{7/2}}{\sqrt{\ell} m^2} \right).$$

The stopped-cost and query gaps are then at least d^δ , and the regret gap is at least $d^{\delta/2}$, up to logarithmic factors.

The stopped-cost comparisons follow by combining the information bounds in Theorem 6.3 with Lemma 2.2, and by truncating the adaptive construction at its first hit. The regret bounds then use the same two-branch argument as (23) and an explore-then-commit use of Theorem 6.5. Full details are in Appendices E and F.

For the full BCO class, closing $\alpha_{\text{full}}^{\text{stop}} \in [5/2, 3]$ requires either uniformly low-cost posterior experiments or a lower bound that rules out all observation-adaptive probes. The dimple result explains why fixed batches, fresh directions, and even adaptive bounded blocks in fresh random subspaces cannot decide that question.

7 Further scope and open problems

7.1 General episodic reinforcement learning

Example 7.1 (Episodic reinforcement learning). *Let M be a fixed unknown MDP with episode length H , rewards in $[0, 1]$, and a common initial-state distribution ρ . One macro-action is a*

complete contingent policy π for an episode, and its observation is the resulting trajectory. After normalizing by H , put

$$\Delta_M(\pi) = \frac{V_M^*(\rho) - V_M^\pi(\rho)}{H} \in [0, 1].$$

At episode boundaries,

$$r(Q) = \inf_{\pi} \mathbb{E}_Q \Delta_M(\pi)$$

satisfies (2), and stopping occurs between episodes. Thus the framework measures episodic pseudo-regret; a realized reward difference need not itself be nonnegative.

The reset to ρ is essential. In continuing control, the physical state changes and $r(Q_t, S_t)$ need not be a supermartingale: the next state can be intrinsically harder without resolving uncertainty. Such problems need a risk-persistence condition or a stateful Bellman formulation such as [Xu \(2026a\)](#).

Corollary 7.2 (Finite episodic models). *Consider any sequence of reset episodic reinforcement learning problems as in Example 7.1, with a finite model class, finite state, action, and reward alphabets, and a finite episode horizon. Using the normalized episode gap,*

$$\alpha_{\text{RL}}^{\text{M}} = \alpha_{\text{RL}}^{\text{B}} = \frac{\alpha_{\text{RL}}^{\text{stop}}}{2}. \quad (89)$$

For fixed H , normalization does not change the dimension exponent. If H grows with d , its dependence must be restored in unnormalized regret.

For preference feedback, the macro-action is a policy pair and the gap is their average normalized value loss, as in (56). Section 4 calculates this object for a finite linear class uniform over a broad action-set sandwich; a known-dynamics chain is one specialization. In a general finite model, the joint latent class of the MDP and comparison kernel must be finite (or satisfy separate compactness and selection conditions). A changing evaluator, deployment in a different MDP, or unknown transitions with reachability constraints requires additional state and a new model-specific stopped experiment.

7.2 Scope beyond paired contextual classes

Section 3.1 proves a contextual result rather than only proposing an analogy: for finite realizable sign classes under iid contexts, it gives a uniform $\log |\mathcal{F}|$ aggregate-feedback certificate and an exact aggregate-versus-semi-bandit separation on balanced product classes. The assumptions are deliberate. The margin ε and context law are known, the latent sign function is fixed, and posterior risk averages over the same stationary context law that generates future rounds.

Large-catalog methods address additional computational and statistical questions. For example, the hierarchy of [Sen et al. \(2021\)](#) makes context-dependent search feasible and observes all or some selected-arm rewards. Extending the exact stopped certificate here to arbitrary reward classes, unknown margins, or adversarial contexts would require a new localization argument; class cardinality alone is insufficient when functions differ only on rare contexts.

Across the applications, the division of labor is constant. The model-specific step is local: construct or rule out a regret-efficient posterior-contraction experiment at one scale. The model-independent transfer then concatenates contractions for upper bounds and stops any regret policy at its first contraction for lower bounds. After checking posterior sufficiency, risk persistence, measurability, and Bayes–minimax closure when needed, a new model requires a new experimental-design argument, not a new regret-conversion proof.

A Proofs for the stopped conversion

Proof of Lemma 2.2. The event $\{t \leq N\}$ lies in $\mathcal{H}_{t-1}$, and on this event the pre-query posterior has not hit the lower-risk set. If π_t is the randomized action kernel, then

$$\mathbb{E}[\Delta_\omega(A_t) \mid \mathcal{H}_{t-1}] = \int \mathbb{E}_{Q_{t-1}} \Delta_\omega(a) \pi_t(da \mid H_{t-1}) \geq r(Q_{t-1}) > \gamma b.$$

Tonelli's theorem applied to the stopped sum proves (6); the comparison follows immediately. $\square$

Proof of Theorem 2.3. If $\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) = +\infty$, the upper statement is the trivial bound T . Otherwise fix one universally measurable selector Σ whose worst-state ratio is at most $\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon$. For the upper bound, use geometric risk buckets $b_j = \vartheta^j$. Whenever $r(Q_t) > 2\eta$, invoke $\Sigma(Q_t, b_j)$ at the unique active scale; otherwise play a measurable δ_t -Bayes action, where $\sum_t \delta_t \leq \delta$. If the horizon cuts through the final experiment block, extend that block only for the purpose of the analysis until its prescribed stopping time; the actual online policy still stops at T . Because every block makes at least one actual query, at most one block is completed in this virtual manner. At scale b_j , the selector's ratio bound gives

$$b_j \mathbb{E}[R_k \mid H_{S_k}] \leq \{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon\} p_k,$$

where p_k is the conditional hit probability. If $\mathcal{K}_j$ is the random set of blocks invoked in bucket j and M_k is the hit indicator for block k , conditional expectation gives the accounting identity

$$\mathbb{E} \sum_{k \in \mathcal{K}_j} p_k = \mathbb{E} \sum_{k \in \mathcal{K}_j} M_k = \mathbb{E}[\text{number of actual hits in bucket } j]. \quad (90)$$

Thus unsuccessful blocks enter the regret sum through their conditional hit probabilities but do not create an additional term.

After the k -th hit in bucket j , at time U_k , one has $r(Q_{U_k}) \leq \gamma b_j$. By (2) and Doob's maximal inequality, applied first over finite horizons after U_k and then by monotone convergence,

$$\mathbb{P} \left\{ \sup_{t \geq U_k} r(Q_t) > \vartheta b_j \mid \mathcal{H}_{U_k} \right\} \leq \frac{\gamma}{\vartheta}.$$

A later return to the same active bucket requires the event inside this probability. The number of hits in one bucket is therefore geometrically dominated and has expectation at most $(1 - \gamma/\vartheta)^{-1}$. Hence

$$\mathbb{E} R_{\text{explore}} \leq \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon}{1 - \gamma/\vartheta} \sum_{b_j \geq 2\eta} \frac{1}{b_j} \leq \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon}{2(1 - \vartheta)(1 - \gamma/\vartheta)\eta}.$$

The non-experiment rounds contribute at most $2T\eta + \delta$. Letting $\delta, \epsilon \downarrow 0$ proves (14); optimizing η , with the boundary choices and the trivial bound T , proves (15).

For the converse, choose an active state with $\text{SC}_b(Q) > c$. Run an arbitrary n -round policy, stop it at $N = \tau_b \wedge n$, and put $p = \mathbb{P}_Q(\tau_b \leq n)$. The stopped-cost premise and nonnegativity give

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \frac{c}{b} p.$$

By the same pre-hit argument as in Lemma 2.2,

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \gamma b \mathbb{E}_Q N \geq \gamma b n (1 - p).$$

Minimizing the maximum of these two affine functions over $p \in [0, 1]$ gives (16). At the displayed calibrated horizon, $n \geq c/(2\gamma b^2)$, and direct substitution gives $\mathbb{E}_Q R_n \geq c/(4b) \geq (\sqrt{\gamma}/4)\sqrt{cn}$. $\square$

Proof of Theorem 2.4. For the lower bound, fix $j \leq J_T$, an active $Q \in \mathcal{S}_j(d)$, and $0 < c < \text{SC}_{b_j, T}(Q)$. Run an arbitrary T -round policy and stop it at $N = \tau_{b_j} \wedge T$. This is a cap- T experiment, so, with $p = \mathbb{P}_Q(\tau_{b_j} \leq T)$,

$$\mathbb{E}_Q R_T \geq \frac{c}{b_j} p, \quad \mathbb{E}_Q R_T \geq \gamma b_j T (1 - p).$$

Minimizing the maximum of these two quantities over $p \in [0, 1]$ gives

$$\mathbb{E}_Q R_T \geq \frac{\gamma b_j T (c/b_j)}{c/b_j + \gamma b_j T}.$$

Let $c \uparrow \text{SC}_{b_j, T}(Q)$, then take the supremum over Q . The displayed harmonic mean is increasing in its second argument, so the supremum replaces $\text{SC}_{b_j, T}(Q)$ by $\mathcal{C}_{j, T}^{\text{pt}}(d)$. Finally, for $x, y \geq 0$,

$$\frac{\gamma xy}{y + \gamma x} \geq \frac{\gamma}{1 + \gamma} \min\{x, y\}.$$

Maximizing over j proves (18); an infinite capped cost is handled by monotone passage in c .

For the upper bound, choose, for each $j \leq J_T$ with $\mathcal{C}_{j, T}^{\text{unif}}(d) < \infty$, a universally measurable cap- T shellwise selector whose worst-state ratio is at most $\mathcal{C}_{j, T}^{\text{unif}}(d) + \epsilon$. At an active state in shell j , use this experiment if

$$\frac{\mathcal{C}_{j, T}^{\text{unif}}(d)}{b_j} \leq b_j T.$$

Otherwise stop experimenting and repeat a fixed measurable δ/T -Bayes action for the rest of the horizon, ignoring subsequent observations. Conditional on the state at which this commitment occurs, its expected remaining regret is at most $b_j T + \delta$. When the risk is at most $1/T$, use a current measurable approximate Bayes action; all such rounds together contribute at most $1 + \delta$.

If the horizon cuts the last invoked block, extend the probability space and simulate only that block to its prescribed cap- T stopping time. The actual prefix regret is no larger, and a post-horizon virtual hit is the final hit and creates no later visit. Thus the conditional-hit accounting and geometric-return argument from Theorem 2.3 remains valid. For every shell on which experimentation is chosen, its expected exploration regret is at most

$$\frac{\mathcal{C}_{j, T}^{\text{unif}}(d) + \epsilon}{b_j(1 - \gamma/\vartheta)}.$$

There can be at most one commitment event. Its cost is the term $b_j T$ for the shell where it occurs; if no commitment occurs, this term is absent. Therefore the total expected regret is bounded by

$$1 + 2\delta + \frac{1}{1 - \gamma/\vartheta} \sum_{j \in \mathcal{E}_T} \frac{\mathcal{C}_{j, T}^{\text{unif}}(d) + \epsilon}{b_j} + \max_{\substack{0 \leq j \leq J_T \\ j \notin \mathcal{E}_T}} b_j T.$$

Let $\delta, \epsilon \downarrow 0$ to obtain (19). Since $(1 - \gamma/\vartheta)^{-1} \geq 1$, the maximum commitment term is at most the same constant times the sum of the commitment terms. This gives (20). The last inequality follows because the sum has $J_T + 1$ terms, each no larger than $\Psi_{\mathfrak{D}}^{\text{unif}}(d, T)$. $\square$

Proof of Theorem 2.6. For $T \leq d^M$, Theorem 2.3 and monotonicity of the scale window give

$$\frac{\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T)}{\sqrt{T}} \leq C_{\gamma, \vartheta} \left\{ T^{-1/2} + \sqrt{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, d^M)} \right\}.$$

Taking exponents proves the upper inequality in (27).

For the reverse inequality, first suppose $\alpha_{\mathfrak{D}}^{\text{stop,pt}} > 0$, and fix

$$0 < q < q' < \alpha_{\mathfrak{D}}^{\text{stop,pt}}.$$

For some fixed m and an infinite sequence of dimensions,

$$1 + \text{SC}_{\mathfrak{D}}^{\text{pt}}(d, d^m) \geq d^{q'}.$$

For all sufficiently large dimensions in this sequence, there is an active state (Q_d, b_d) , with $b_d \geq d^{-m}$, such that

$$\text{SC}_{b_d}(Q_d) > c_d, \quad c_d = d^q.$$

Set

$$n_d = \left\lfloor \frac{c_d}{\gamma b_d^2} \right\rfloor.$$

Then $c_d/(\gamma b_d^2) \geq 2$ for large d , and

$$n_d \leq \gamma^{-1} d^{q+2m} \leq d^M$$

for one fixed integer M . The calibrated lower bound in Theorem 2.3 yields

$$\frac{\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, n_d)}{\sqrt{n_d}} \geq \frac{\sqrt{\gamma}}{4} \sqrt{c_d} = \frac{\sqrt{\gamma}}{4} d^{q/2}.$$

Thus $\alpha_{\mathfrak{D}}^{\text{B}} \geq q/2$. Let $q \uparrow \alpha_{\mathfrak{D}}^{\text{stop,pt}}$; if the latter exponent is infinite, take arbitrary finite q . When it is zero, the lower inequality is immediate.

The measurable-choice assumption identifies the two finite stopped values, and hence their exponents. Bayes–minimax equality identifies the remaining regret exponents. $\square$

Proof of Corollary 2.7. Write $m = |\Omega_d|$. For each $a \in \mathcal{A}_d$, the measure

$$\lambda_a = \frac{1}{m} \sum_{\omega \in \Omega_d} P_{\omega}(\cdot \mid a)$$

dominates all $P_{\omega}(\cdot \mid a)$. Choose Borel Radon–Nikodym derivatives $p_{\omega,a}$. Off a predictive-null set, the posterior update is

$$\Psi(Q, a, o)(\omega) = \frac{Q(\omega) p_{\omega,a}(o)}{\sum_{\omega'} Q(\omega') p_{\omega',a}(o)},$$

with an arbitrary fixed value when the denominator is zero. Hence the belief transition is Borel. Moreover,

$$r(Q) = \min_{a \in \mathcal{A}_d} \sum_{\omega \in \Omega_d} Q(\omega) \Delta_{\omega}(a)$$

is Borel, and deterministic tie-breaking selects a Bayes action measurably.

For every finite cap L , the penalized stopped Bellman recursion (7) therefore has Borel primitives and a finite action set. Finite-horizon measurable selection supplies a measurable near-minimizer. The rational-price argument in (8) then gives a universally measurable ϵ -optimal selector for the capped ratio. Finally, choose the first cap L whose value lies within $\epsilon/2$ of

$$\text{SC}_b(Q) = \inf_{L \geq 1} \text{SC}_{b,L}(Q).$$

This gives an ϵ -optimal selector for the uncapped stopped cost, so the pointwise and uniform profiles coincide.

For Bayes–minimax equality, let

$$\mathcal{V}_T = \left\{ (\mathbb{E}_\omega^\pi R_T)_{\omega \in \Omega_d} : \pi \right\} \subset [0, T]^m.$$

Independent randomization at H_0 makes $\mathcal{V}_T$ convex. The prior simplex $\mathcal{P}(\Omega_d)$ is compact, so Sion’s minimax theorem gives

$$\inf_{v \in \mathcal{V}_T} \max_{\omega} v_{\omega} = \sup_{Q \in \mathcal{P}(\Omega_d)} \inf_{v \in \mathcal{V}_T} \langle Q, v \rangle.$$

The two sides are frequentist minimax and worst-prior Bayes regret, respectively. The exponent identity now follows from Theorem 2.6. $\square$

Proof of Corollary 7.2. The latent MDP class and the complete contingent-policy set are finite, as is the trajectory observation space. Apply Corollary 2.7 to the episode-level problem. $\square$

B Proofs for aggregate-feedback top- k bandits

Proof of Proposition 3.3. We first prove the upper bound for arbitrary k , without assuming the existence of a Hadamard matrix. For

$$n_0 = \lceil C_\gamma k / \varepsilon^2 \rceil,$$

there is a deterministic matrix $A \in \{-1, +1\}^{n_0 \times k}$ such that

$$A^\top A \succeq \frac{n_0}{2} I_k. \tag{91}$$

Indeed, take independent Rademacher rows. For a fixed unit vector u , $\langle A_t, u \rangle$ is isotropic sub-Gaussian, so its empirical second moment differs from one by at most $1/4$ with probability $1 - 2e^{-cn_0}$. A $1/8$ -net of the unit sphere has size at most 17^k ; a union bound followed by the standard net extension gives (91) when $n_0 \geq Ck$. Increasing C_γ covers all $0 < \varepsilon \leq 1/4$.

Every row a_t is a feasible top- k query: choose $(i, +)$ when $a_{t,i} = 1$ and $(i, -)$ when $a_{t,i} = -1$. Centering the observations in (34) gives

$$Z = \varepsilon A \theta + \xi, \quad \xi \sim N(0, k I_{n_0}).$$

The least-squares estimator

$$\hat{v} = \frac{1}{\varepsilon} (A^\top A)^{-1} A^\top Z$$

satisfies

$$\hat{v} - \theta \sim N\left(0, \frac{k}{\varepsilon^2} (A^\top A)^{-1}\right), \quad \text{Var}(\hat{v}_i - \theta_i) \leq \frac{2k}{\varepsilon^2 n_0} \leq \frac{2}{C_\gamma}.$$

Let $\hat{\theta}_i = \text{sign}(\hat{v}_i)$, with an arbitrary convention at zero. A Gaussian tail bound gives

$$\mathbb{P}(\hat{\theta}_i \neq \theta_i) \leq e^{-C_\gamma/4}.$$

Choose C_γ so that the right-hand side is at most $\gamma/16$.

Conditional on the transcript, the coordinatewise posterior sign rule has Hamming risk

$$\frac{1}{2} \sum_{i=1}^k (1 - |m_i|),$$

and is no worse than any transcript-measurable estimator. Consequently,

$$\mathbb{E} \sum_{i=1}^k (1 - |m_i|) \leq 2 \mathbb{E} d_H(\hat{\theta}, \theta) \leq \frac{\gamma k}{8}.$$

Markov's inequality shows that the terminal posterior risk in (36) is at most $\gamma\varepsilon$ with probability at least $7/8$.

Run this predetermined design, but stop at the first contraction. Since all gaps are nonnegative, first-hit truncation can only reduce its regret. For every predetermined row a_t , symmetry of the product prior gives

$$\mathbb{E} \Delta_\theta(S_t) = \varepsilon \left\{ 1 - \frac{1}{k} \mathbb{E} \langle a_t, \theta \rangle \right\} = \varepsilon.$$

Therefore, with $p \geq 7/8$,

$$\frac{\varepsilon \mathbb{E} R_N}{p} \leq \frac{8}{7} \varepsilon^2 n_0 \leq C_\gamma k,$$

after enlarging the constant to cover the ceiling and $k = 1$.

For the reverse inequality, Lemma B.1 applied to (34) gives, for every admissible experiment with success probability p ,

$$\mathbb{E} N \geq c_\gamma \frac{pk}{\varepsilon^2}.$$

Lemma 2.2 now yields

$$\frac{\varepsilon \mathbb{E} R_N}{p} \geq \gamma \varepsilon^2 \frac{\mathbb{E} N}{p} \geq c_\gamma k.$$

□

Proof of Theorem 3.1. The upper bound is the minimum of the trivial kT bound and the distribution-free full-bandit guarantee $O(k\sqrt{nT \log(eT)})$ of [Rejwan and Mansour \(2020\)](#), specialized to $n = 2k$. Their guarantee is stated for independent 1-sub-Gaussian arms, which includes the present unit-variance Gaussian model.

For the lower bound, fix a constant $0 < c_0 < c_\gamma$, where $c_\gamma k$ is the lower stopped cost in Proposition 3.3. If $T \geq Ck$, choose $\varepsilon = c\sqrt{k/T}$ with $c > 0$ small enough that $\varepsilon \leq 1/4$. Apply Theorem 2.3 with $B \geq \varepsilon^{-1}$, the paired prior Q_0 , and stopped-cost threshold $c_0 k$. Equation (16) gives

$$\sup_Q \inf_\pi \mathbb{E}_Q R_T \geq \frac{1}{2} \min \left\{ \gamma \varepsilon T, \frac{c_0 k}{\varepsilon} \right\} \geq c' \sqrt{kT}$$

in normalized units. The frequentist minimax value dominates this Bayes value, and multiplication by k restores sum-reward units.

If $T < Ck$, take a sufficiently small constant $\varepsilon \leq 1/4$. The same display gives $\Omega(T)$ normalized regret (with constants adjusted on the overlap between the two ranges), and hence $\Omega(kT)$ in raw units. The case $T = 1$ is included. □

Proof of Corollary 3.4. The upper bound combines the trivial bound kT with the guarantee

$$O\left(k\sqrt{nT\log(eT)}\right)$$

of [Rejwan and Mansour \(2020\)](#). For the lower bound, restrict the n -arm model to the following subclass. Keep the $2k$ paired arms in (33), and let the remaining $n - 2k$ arms be known mean-zero, unit-variance Gaussian decoys. If a query S contains q_S decoys, then, after subtracting the known offset,

$$Y - \frac{k - q_S}{2} = \varepsilon \langle a(S), \theta \rangle + \xi, \quad \xi \sim N(0, k),$$

where $\|a(S)\|_2^2 \leq k - q_S \leq k$, and its raw gap is

$$\Delta_\theta^{\text{raw}}(S) = \frac{q_S}{2} + \varepsilon \{k - \langle a(S), \theta \rangle\}.$$

For every posterior on θ , the posterior-preferred arm in each pair has mean at least $1/2$, whereas every decoy has mean zero. Hence a Bayes action still chooses one posterior-preferred arm from every pair, and posterior risk remains (36).

The coordinate-flip calculation in the proof of Proposition 3.3 is unchanged: the offset contains no information about θ , the observation variance is still k , and $\|a(S)\|_2^2 \leq k$. Thus every stopped experiment that contracts risk with probability p satisfies

$$\mathbb{E}N \geq c_\gamma \frac{pk}{\varepsilon^2}.$$

The additional term $q_S/2$ in the gap is nonnegative, so the same pre-hit regret argument gives $\text{SC}_\varepsilon(Q_0) \geq c_\gamma k$. Applying the lower half of Theorem 2.3 and tuning ε exactly as in the proof of Theorem 3.1 yields

$$\mathfrak{R}_{n,k}^*(T) \geq c \min\{kT, k^{3/2}\sqrt{T}\}.$$

This proves (39). Finally, if $2k \leq n \leq C_0 k$, then $k\sqrt{nT} \leq \sqrt{C_0} k^{3/2}\sqrt{T}$. Combining this inequality with (39) gives the asserted $\tilde{\Theta}_{C_0}$ characterization. $\square$

B.1 A stopped direct-product lemma

Lemma B.1 (Stopped Assouad bound for Gaussian sums). *Let θ be uniform on $\{-1, +1\}^k$. At time t , an adaptive policy chooses $a_t \in \{-1, 0, 1\}^k$, with $\|a_t\|_2^2 \leq k$, and observes*

$$Y_t = \varepsilon \langle a_t, \theta \rangle + \xi_t, \quad \xi_t \stackrel{\text{iid}}{\sim} N(0, k).$$

Let N be the first time that $\sum_i (1 - |\mathbb{E}[\theta_i | H_N]|) \leq \gamma k$, or an earlier voluntary stopping time, and let $\mathbf{M}$ be the first alternative. If $p = \mathbb{P}(\mathbf{M})$, then

$$\mathbb{E}N \geq c_\gamma \frac{pk}{\varepsilon^2}.$$

Proof. For $i \in [k]$, pair each parameter θ with $\theta^{(i)}$, obtained by flipping its i -th coordinate, and run the two transcript laws with the same policy kernel. The stopped Gaussian chain rule gives

$$D_{\text{KL}}(P_\theta^N \| P_{\theta^{(i)}}^N) = \frac{2\varepsilon^2}{k} \mathbb{E}_\theta \sum_{t \leq N} a_{t,i}^2. \quad (92)$$

If $\mathbb{E}N = \infty$, the desired duration lower bound is immediate. Otherwise the identity follows first for $N \wedge L$, then from convergence of relative entropy along the increasing stopped-history σ -fields and Tonelli's theorem for the nonnegative conditional KL increments. Averaging over θ , summing over i , and using $\sum_i a_{t,i}^2 \leq k$ yields

$$2^{-k} \sum_{\theta} \sum_{i=1}^k D_{\text{KL}}(P_{\theta}^N \| P_{\theta(i)}^N) \leq 2\epsilon^2 \mathbb{E}N. \quad (93)$$

For each i , let P_i^+ and P_i^- be the transcript laws conditional on $\theta_i = +1$ and $\theta_i = -1$, respectively. Joint convexity of relative entropy and the pairing above give

$$\sum_i \{D_{\text{KL}}(P_i^+ \| P_i^-) + D_{\text{KL}}(P_i^- \| P_i^+)\} \leq 2^{1-k} \sum_{\theta} \sum_i D_{\text{KL}}(P_{\theta}^N \| P_{\theta(i)}^N).$$

Let $P = (P_i^+ + P_i^-)/2$, which is the same unconditional transcript law for every i . Bayes' rule gives

$$m_i(h) = \frac{dP_i^+ - dP_i^-}{dP_i^+ + dP_i^-}(h).$$

Consequently the i -th Jeffreys divergence is

$$D_{\text{KL}}(P_i^+ \| P_i^-) + D_{\text{KL}}(P_i^- \| P_i^+) = 4\mathbb{E}_P[m_i(H_N) \operatorname{arctanh} m_i(H_N)].$$

Since $x \operatorname{arctanh} x \geq x^2$, if a stopped-transcript event M satisfies

$$\sum_i (1 - |\mathbb{E}[\theta_i | H_N]|) \leq \gamma k \quad \text{on } M,$$

then Cauchy–Schwarz yields, pointwise on M ,

$$\sum_i m_i(H_N)^2 \geq k^{-1} \left(\sum_i |m_i(H_N)| \right)^2 \geq (1 - \gamma)^2 k.$$

It follows that

$$2^{-k} \sum_{\theta} \sum_{i=1}^k D_{\text{KL}}(P_{\theta}^N \| P_{\theta(i)}^N) \geq 2(1 - \gamma)^2 k \mathbb{P}(M). \quad (94)$$

Take $M = \mathcal{M}$ and combine (93)–(94). □

C Proofs for finite contextual aggregate feedback

Throughout this section, regret is normalized by k , as in the general theory. We restore raw selected-sum units when deriving the regret comparisons.

C.1 A finite-class random-design bound

For probability measures P, P' , write

$$\rho_{\text{H}}(P, P') = \int \sqrt{dP dP'}$$

for their Hellinger affinity.

Lemma C.1 (Random aggregate design). *Let A be uniform on $\{-1, +1\}^k$, independently of $X \sim \nu$, and let*

$$Z = \varepsilon \langle A, f(X) \rangle + \xi, \quad \xi \sim N(0, k).$$

For

$$h_\nu(f, g) = \mathbb{E}_{X \sim \nu} d_H(f(X), g(X)),$$

the one-sample laws P_f and P_g of (X, A, Z) satisfy

$$\rho_H(P_f, P_g) \leq \exp \left\{ -c \frac{\varepsilon^2}{k} h_\nu(f, g) \right\} \quad (95)$$

for a universal $c > 0$.

Consequently, if $\hat{f}_N$ is a maximum-likelihood estimator over a finite class $\mathcal{F}$ from N independent random-design samples, with deterministic tie-breaking, then

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f h_\nu(f, \hat{f}_N) \leq C \min \left\{ k, \frac{k \log(e|\mathcal{F}|)}{N \varepsilon^2} \right\}. \quad (96)$$

Proof. Conditional on $X = x$ and $A = a$, the two observation laws are Gaussian with common variance k and mean difference

$$\varepsilon \langle a, f(x) - g(x) \rangle.$$

The affinity of two Gaussians with common variance k and mean difference u is $\exp\{-u^2/(8k)\}$. Hence

$$\rho_H(P_f, P_g) = \mathbb{E}_{X, A} \exp \left\{ -\frac{\varepsilon^2 \langle A, f(X) - g(X) \rangle^2}{8k} \right\}.$$

Fix x , put $r = d_H(f(x), g(x))$, and write

$$B = \frac{1}{2} \langle A, f(x) - g(x) \rangle.$$

Up to deterministic signs, B is a sum of r independent Rademacher variables. Thus

$$\mathbb{E} B^2 = r, \quad \mathbb{E} B^4 = 3r^2 - 2r \leq 3r^2.$$

For $r \geq 1$, Paley–Zygmund gives

$$\mathbb{P}\{B^2 \geq r/2\} \geq \frac{1}{12}.$$

Because $r \leq k$ and $\varepsilon \leq 1/4$,

$$\frac{\varepsilon^2 r}{4k} \leq \frac{1}{64}.$$

It follows that

$$\mathbb{E}_A \left[1 - \exp \left\{ -\frac{\varepsilon^2 B^2}{2k} \right\} \right] \geq c \frac{\varepsilon^2 r}{k}.$$

The same inequality is trivial for $r = 0$. Averaging over X yields

$$1 - \rho_H(P_f, P_g) \geq c \frac{\varepsilon^2}{k} h_\nu(f, g),$$

and $1 - u \leq e^{-u}$ proves (95).

For the second claim, if $\hat{f}_N = g$, then the likelihood of g is at least that of the truth f . Markov's inequality applied to the square root of the likelihood ratio gives

$$\mathbb{P}_f(\hat{f}_N = g) \leq \rho_H(P_f^{\otimes N}, P_g^{\otimes N}) \leq \exp \left\{ -c \frac{N\varepsilon^2}{k} h_\nu(f, g) \right\}.$$

Therefore, for every $u \geq 0$,

$$\mathbb{P}_f\{h_\nu(f, \hat{f}_N) \geq u\} \leq \min \left\{ 1, |\mathcal{F}| \exp \left\{ -c \frac{N\varepsilon^2}{k} u \right\} \right\}.$$

Integrating this tail bound and using $h_\nu \leq k$ gives (96). $\square$

Proof of Proposition 3.5. Fix a prior Q and an active scale b . Consider L future queries that, after observing the context, independently choose A_t uniformly from $\{-1, +1\}^k$ and select arm $(i, A_{t,i})$ from every pair. These are feasible grouped k -subsets.

Let $\hat{f}_L$ be the maximum-likelihood estimator from the resulting L aggregate observations. For every $f, g \in \mathcal{F}$, playing the contextual rule prescribed by g under truth f has normalized expected gap

$$\frac{\varepsilon}{k} \mathbb{E}_X \{k - \langle g(X), f(X) \rangle\} = \frac{2\varepsilon}{k} h_\nu(f, g). \quad (97)$$

Since a posterior Bayes action is no worse than the action prescribed by $\hat{f}_L$,

$$r(Q_L) \leq \mathbb{E}_{f \sim Q_L} \frac{2\varepsilon}{k} h_\nu(f, \hat{f}_L).$$

Taking expectation over $f \sim Q$ and the future transcript, and then applying Lemma C.1, gives

$$\mathbb{E}_Q r(Q_L) \leq C \frac{\log(e|\mathcal{F}|)}{L\varepsilon}. \quad (98)$$

Choose

$$L = \left\lceil C_\gamma \frac{\log(e|\mathcal{F}|)}{\varepsilon b} \right\rceil$$

with C_γ sufficiently large. Equation (98) and Markov's inequality imply

$$\mathbb{P}_Q\{r(Q_L) \leq \gamma b\} \geq \frac{7}{8}.$$

Run this design but truncate it at the first hit of $\{r \leq \gamma b\}$. Its success probability p is at least $7/8$.

For every truth f , context x , and random grouped query A ,

$$\mathbb{E}_A \left[\frac{\varepsilon}{k} \{k - \langle A, f(x) \rangle\} \right] = \varepsilon.$$

First-hit truncation and nonnegativity therefore give

$$\mathbb{E}_Q R_N \leq \varepsilon L.$$

Consequently,

$$\frac{b \mathbb{E}_Q R_N}{p} \leq \frac{8}{7} b \varepsilon L \leq C_\gamma \log(e|\mathcal{F}|),$$

where the ceiling is absorbed using $b \leq 1$, $\varepsilon \leq 1/4$, and $\log(e|\mathcal{F}|) \geq 1$. This proves (44). The experiment depends on (Q, b) only through the measurable integer L ; all remaining tie-breaking and randomization can be fixed measurably.

For completeness, the regret consequence also follows directly and does not require an efficient maximum-likelihood oracle. Run the random aggregate design for

$$L_T = \min \left\{ T, \left\lceil \frac{\sqrt{T \log(e|\mathcal{F}|)}}{\varepsilon} \right\rceil \right\}$$

rounds, and, if $L_T < T$, commit to the contextual rule prescribed by $\hat{f}_{L_T}$. Exploration has expected raw regret $k\varepsilon L_T$. By (97) and (96), exploitation contributes at most

$$\frac{CkT \log(e|\mathcal{F}|)}{L_T \varepsilon}.$$

The definition of L_T , including the case $L_T = T$, therefore gives raw regret at most

$$Ck \left\{ 1 + \sqrt{T \log(e|\mathcal{F}|)} \right\}.$$

Independently playing uniform Rademacher grouped queries throughout has expected raw regret exactly $k\varepsilon T$, uniformly over $f \in \mathcal{F}$. Taking the better policy proves (45). $\square$

C.2 The balanced product class

In this subsection, let

$$d = qk, \quad \theta = (\theta_{x,i})_{x \in [q], i \in [k]} \sim \text{Unif}\{-1, +1\}^{q \times k}.$$

For a stopped transcript, write

$$m_{x,i}(H_N) = \mathbb{E}[\theta_{x,i} \mid H_N].$$

By (43), success at scale $b = \varepsilon$ implies

$$\sum_{x=1}^q \sum_{i=1}^k \{1 - |m_{x,i}(H_N)|\} \leq \gamma d. \quad (99)$$

Proof of the aggregate part of Proposition 3.6. We first prove the lower bound. For $u = (x, i)$, let $\theta^{(u)}$ be obtained from θ by flipping $\theta_{x,i}$. Under aggregate feedback, the stopped Gaussian chain rule gives

$$D_{\text{KL}}(P_{\theta}^N \| P_{\theta^{(x,i)}}^N) = \frac{2\varepsilon^2}{k} \mathbb{E}_{\theta} \sum_{t \leq N} \mathbf{1}\{X_t = x\} a_{t,i}^2. \quad (100)$$

Indeed, when $X_t = x$, flipping $\theta_{x,i}$ changes the conditional Gaussian mean by $2\varepsilon a_{t,i}$; when $X_t \neq x$, it does not change the observation law. If $\mathbb{E}N = \infty$, the desired duration bound is immediate. Otherwise, (100) follows first for $N \wedge L$ and then by convergence of relative entropy along the increasing stopped history sigma-fields, together with Tonelli's theorem for the nonnegative conditional KL increments.

Averaging over θ , summing over x, i , and using $\sum_i a_{t,i}^2 \leq k$, we obtain

$$2^{-d} \sum_{\theta} \sum_{x,i} D_{\text{KL}}(P_{\theta}^N \| P_{\theta^{(x,i)}}^N) \leq 2\varepsilon^2 \mathbb{E}N. \quad (101)$$

There is no additional factor $1/q$: a fixed bit is observed only when its context occurs, but summing over the q context blocks removes that frequency factor.

For each $u = (x, i)$, let P_u^+ and P_u^- denote the stopped transcript laws conditional on $\theta_u = +1$ and $\theta_u = -1$. Joint convexity of relative entropy gives

$$\sum_u \{D_{\text{KL}}(P_u^+ \| P_u^-) + D_{\text{KL}}(P_u^- \| P_u^+)\} \leq 2^{1-d} \sum_{\theta} \sum_u D_{\text{KL}}(P_{\theta}^N \| P_{\theta(u)}^N).$$

Let $P = (P_u^+ + P_u^-)/2$, which is the same unconditional stopped transcript law for every u . Bayes' rule gives

$$\frac{dP_u^+}{dP} = 1 + m_u, \quad \frac{dP_u^-}{dP} = 1 - m_u,$$

and hence

$$D_{\text{KL}}(P_u^+ \| P_u^-) + D_{\text{KL}}(P_u^- \| P_u^+) = 4\mathbb{E}_P[m_u \operatorname{arctanh}(m_u)] \geq 4\mathbb{E}_P m_u^2.$$

On the success event M , (99) and Cauchy–Schwarz imply

$$\sum_u m_u^2 \geq \frac{1}{d} \left(\sum_u |m_u| \right)^2 \geq (1 - \gamma)^2 d.$$

Writing $p = \mathbb{P}(M)$, we conclude that

$$2^{-d} \sum_{\theta} \sum_u D_{\text{KL}}(P_{\theta}^N \| P_{\theta(u)}^N) \geq 2(1 - \gamma)^2 dp. \quad (102)$$

Combining (101) and (102) gives

$$\mathbb{E}N \geq (1 - \gamma)^2 \frac{p q k}{\varepsilon^2}.$$

Lemma 2.2, applied at $b = \varepsilon$, yields

$$\frac{\varepsilon \mathbb{E}R_N}{p} \geq \gamma \varepsilon^2 \frac{\mathbb{E}N}{p} \geq c_{\gamma} q k.$$

Taking the infimum over stopped experiments proves the lower bound in (47).

For the upper bound, run

$$N_0 = \left\lceil C_{\gamma} \frac{q k}{\varepsilon^2} \right\rceil$$

independent random grouped queries $A_t \in \{-1, +1\}^k$. Put $Z_t = Y_t - k/2$ and define

$$\hat{v}_{x,i} = \frac{q}{\varepsilon N_0} \sum_{t=1}^{N_0} \mathbf{1}\{X_t = x\} A_{t,i} Z_t. \quad (103)$$

Under truth θ ,

$$\mathbb{E}_{\theta}[\mathbf{1}\{X_t = x\} A_{t,i} Z_t] = \frac{\varepsilon}{q} \theta_{x,i}.$$

Moreover,

$$\mathbb{E}_{\theta}[\{\mathbf{1}\{X_t = x\} A_{t,i} Z_t\}^2] = \frac{k(1 + \varepsilon^2)}{q}.$$

Therefore

$$\mathbb{E}_{\theta}(\hat{v}_{x,i} - \theta_{x,i})^2 \leq \frac{q k (1 + \varepsilon^2)}{\varepsilon^2 N_0}. \quad (104)$$

Choose C_γ so that the right-hand side is at most $\gamma/32$. With $\hat{\theta}_{x,i} = \text{sign}(\hat{v}_{x,i})$, Chebyshev's inequality gives

$$\mathbb{E}d_H(\hat{\theta}, \theta) \leq \frac{\gamma d}{32}.$$

Conditional on the transcript, the coordinatewise posterior sign rule is the Bayes estimator for Hamming loss. Hence

$$\mathbb{E} \sum_{x,i} (1 - |m_{x,i}|) \leq 2\mathbb{E}d_H(\hat{\theta}, \theta) \leq \frac{\gamma d}{16}.$$

Markov's inequality shows that (99) holds with probability at least $15/16$.

Run the design only until its first hit. Every random grouped query has prior expected normalized gap ε , so

$$\frac{\varepsilon \mathbb{E}R_N}{\mathbb{P}(\mathbf{M})} \leq C_\gamma qk.$$

This proves the upper bound in (47). $\square$

Proof of the semi-bandit part of Proposition 3.6. Consider first a stopped semi-bandit experiment with success probability p . Let

$$c_{t,i} = \mathbf{1}\{(i, +1) \in S_t\} + \mathbf{1}\{(i, -1) \in S_t\}.$$

Thus $c_{t,i} \in \{0, 1, 2\}$ and $\sum_i c_{t,i} = k$. Flipping $\theta_{x,i}$ changes the mean of each observed arm from pair i by magnitude 2ε . The stopped chain rule therefore gives

$$D_{\text{KL}}(P_\theta^N \| P_{\theta(x,i)}^N) = 2\varepsilon^2 \mathbb{E}_\theta \sum_{t \leq N} \mathbf{1}\{X_t = x\} c_{t,i}.$$

Averaging and summing over the qk flips yields

$$2^{-d} \sum_\theta \sum_{x,i} D_{\text{KL}}(P_\theta^N \| P_{\theta(x,i)}^N) \leq 2\varepsilon^2 k \mathbb{E}N. \quad (105)$$

The localized lower bound (102) depends only on the posterior means and the success event, not on the feedback mechanism. Combining it with (105) gives

$$\mathbb{E}N \geq c_\gamma \frac{pq}{\varepsilon^2}.$$

Lemma 2.2 now implies

$$\frac{\varepsilon \mathbb{E}R_N}{p} \geq c_\gamma q.$$

For the reverse bound, at each realized context choose independent Rademacher signs $A_{t,i}$, select arm $(i, A_{t,i})$ from each pair, and observe its individual reward $W_{t,i}$. For

$$N_0 = \lceil C_\gamma q / \varepsilon^2 \rceil,$$

define

$$\hat{v}_{x,i}^{\text{semi}} = \frac{q}{\varepsilon N_0} \sum_{t=1}^{N_0} \mathbf{1}\{X_t = x\} A_{t,i} (W_{t,i} - 1/2).$$

Then

$$\mathbb{E}_\theta \hat{v}_{x,i}^{\text{semi}} = \theta_{x,i}, \quad \mathbb{E}_\theta (\hat{v}_{x,i}^{\text{semi}} - \theta_{x,i})^2 \leq \frac{q(1 + \varepsilon^2)}{\varepsilon^2 N_0}.$$

The same posterior-sign and Markov argument used above gives constant-probability contraction. Every such query again has prior expected normalized gap ε . Consequently,

$$\text{SC}_\varepsilon^{\text{semi}}(Q_0) \leq C_\gamma q.$$

Together with the lower bound, this proves (48). $\square$

C.3 Regret comparisons

Proof of (49) and (50). The lower bounds follow directly from Theorem 2.3 at $b = \varepsilon$ and Proposition 3.6. Restoring raw units,

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T) \geq c_\gamma k \min \left\{ \varepsilon T, \frac{qk}{\varepsilon} \right\}, \quad (106)$$

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{semi}}(T) \geq c_\gamma k \min \left\{ \varepsilon T, \frac{q}{\varepsilon} \right\}. \quad (107)$$

For aggregate feedback, split exploration into $L = \lceil C \log(eT) \rceil$ independent blocks, each of length

$$n_{\text{agg}} = \lceil C_\gamma qk / \varepsilon^2 \rceil.$$

Apply (103) separately in each block and take the coordinatewise majority of the L estimated signs. After increasing C , each final coordinate has error probability at most $(eT)^{-4}$. If $Ln_{\text{agg}} \leq T$, commit to these signs for the remaining rounds. The expected raw exploration regret is at most

$$\varepsilon k L n_{\text{agg}} \leq C_\gamma \frac{k(qk) \log(eT)}{\varepsilon},$$

and the expected exploitation regret is at most a universal constant. If $Ln_{\text{agg}} > T$, play independent uniform Rademacher grouped queries throughout, at expected raw regret $\varepsilon k T$. This proves (49).

Semi-bandit feedback admits a log-free online policy because the symmetric baseline $1/2$ is known. For each context-coordinate pair (x, i) , initialize a score $S_{x,i} = 0$. At the n -th occurrence of context x , select from pair i the arm with sign

$$A_{x,i,n} = \text{sign}(S_{x,i}),$$

breaking a zero tie uniformly, observe its reward $W_{x,i,n}$, and update

$$S_{x,i} \leftarrow S_{x,i} + A_{x,i,n}(W_{x,i,n} - 1/2).$$

Conditional on the past,

$$A_{x,i,n}(W_{x,i,n} - 1/2) = \varepsilon \theta_{x,i} + \xi'_{x,i,n}, \quad \xi'_{x,i,n} \sim N(0, 1),$$

and these transformed noises are independent. Therefore

$$\mathbb{P}_\theta \{A_{x,i,n} \neq \theta_{x,i}\} \leq \exp\{-\varepsilon^2(n-1)/2\}.$$

An error on one pair costs 2ε in raw units. Extending the sum beyond the random number of visits to each context gives

$$\mathbb{E}_\theta R_T^{\text{semi}} \leq 2\varepsilon \sum_{x=1}^q \sum_{i=1}^k \sum_{n \geq 1} \exp\{-\varepsilon^2(n-1)/2\} \leq C \frac{qk}{\varepsilon}.$$

Uniform random grouped play also gives $\mathbb{E} R_T^{\text{semi}} = \varepsilon k T$. Taking the better policy and combining with (107) proves the Θ statement in (50). $\square$

Same-scale and worst-scale comparisons. Put $d = qk$. Suppose first that $T \geq 16d$ and take $\varepsilon = c\sqrt{d/T}$ for a fixed sufficiently small $c > 0$. The aggregate lower bound (106) and the trivial upper bound $2k\varepsilon T$ give

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T) = \Theta(k\sqrt{dT}).$$

The semi-bandit lower bound (107) and the online policy above give

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{semi}}(T) = \Theta(\sqrt{dT}).$$

Thus the same horizon-calibrated class has an exact factor- k separation.

For every $A \geq 1$,

$$\sup_{0 < \varepsilon \leq 1/4} \min \left\{ \varepsilon T, \frac{A}{\varepsilon} \right\} = \Theta \left(\min\{T, \sqrt{AT}\} \right). \quad (108)$$

The lower direction chooses $\varepsilon = \sqrt{A/T}$ when $T \geq 16A$ and $\varepsilon = 1/4$ otherwise; the upper direction is elementary. Using $A = qk$ in (106) gives the lower bound in (51). Its upper bound follows from Proposition 3.5 with $|\mathcal{F}_{q,k}| = 2^{qk}$. Using $A = q$ in (107), together with the log-free semi-bandit policy above, proves (52). Their ratio in the common square-root regime $T \gtrsim qk$ is $\Theta(\sqrt{k})$. $\square$

Proof of (53). The upper bound follows immediately from Proposition 3.5. For every $|\mathcal{F}| \leq M$,

$$\mathfrak{R}_{\mathcal{F},\varepsilon}^{\text{agg}}(T) \leq Ck \min \left\{ T, 1 + \sqrt{T \log(eM)} \right\}.$$

Taking the supremum over the admissible contextual problems proves the right-hand side of (53).

For the lower bound, let

$$D = \log_2 M, \quad q = \left\lfloor \frac{D}{k} \right\rfloor, \quad d = qk.$$

The assumption $M \geq 2^k$ ensures $q \geq 1$. Moreover,

$$\frac{D}{2} \leq d \leq D:$$

if $D < 2k$, then $q = 1$ and $d = k > D/2$; if $D \geq 2k$, then $d \geq D - k \geq D/2$. The balanced class $\mathcal{F}_{q,k}$ has cardinality $2^d \leq M$.

If $T \geq 16d$, choose $\varepsilon = \sqrt{d/T} \leq 1/4$. The aggregate lower bound (106) gives

$$\mathfrak{R}_{q,k,\varepsilon}^{\text{agg}}(T) \geq ck\sqrt{dT}.$$

If $T < 16d$, choose $\varepsilon = 1/4$; then

$$\mathfrak{R}_{q,k,1/4}^{\text{agg}}(T) \geq ckT.$$

Since d and $\log M$ differ by at most universal constant factors, the two cases imply

$$\mathfrak{R}_{k,M}^{\text{agg}}(T) \geq ck \min\{T, \sqrt{T \log M}\}.$$

This proves the left-hand side of (53). $\square$

D Proofs for linear and combinatorial preferences

Proof of Proposition 4.1. Write

$$\ell_F(u) = 2F(u) - 1.$$

For $F = \Phi$ and $F = \sigma$, ℓ_F is odd and increasing.

Upper stopped bound. Put

$$\alpha = \frac{\beta b}{d}.$$

Define

$$\rho_d(\alpha) = \mathbb{E} \left[W_1 \ell_F \left(2\alpha \sum_{j=1}^d W_j \right) \right], \quad W_1, \dots, W_d \stackrel{\text{iid}}{\sim} \text{Unif}\{-1, +1\}.$$

There is a constant $c_F > 0$ such that, for every $d \geq 1$ and $\alpha > 0$,

$$\rho_d(\alpha) \geq c_F \min\{\alpha, d^{-1/2}\}, \quad \text{and hence} \quad \rho_d(\alpha)^2 \geq \frac{c_F}{d + \alpha^{-2}} = \frac{c_F}{D_{d,b,\beta}}. \quad (109)$$

Indeed, conditioning on $S = \sum_{j=2}^d W_j$ and pairing the two values of W_1 gives

$$\rho_d(\alpha) = \frac{1}{2} \mathbb{E} [\ell_F\{2\alpha(S+1)\} - \ell_F\{2\alpha(S-1)\}].$$

For $\alpha \leq 1/8$, restrict to $|S| \leq (8\alpha)^{-1}$. Both arguments then lie in a fixed compact interval, on which ℓ'_F is bounded below. A central-binomial estimate gives, for every $L \geq 1$,

$$\mathbb{P}\{|S| \leq L\} \geq c \min\{1, L/\sqrt{d}\},$$

with the interval adjusted by one lattice unit if necessary. Therefore

$$\rho_d(\alpha) \geq c_F \alpha \min\{1, (\alpha\sqrt{d})^{-1}\} = c_F \min\{\alpha, d^{-1/2}\}.$$

For $\alpha > 1/8$, restrict instead to the central atom $S = 0$ when $d-1$ is even, and to $S \in \{-1, +1\}$ when $d-1$ is odd. Its probability is at least $c/\sqrt{d}$, while the corresponding difference of the two ℓ_F -values is bounded below by a positive F -dependent constant. This proves (109).

At each query draw $\xi \in \{-1, +1\}^d$ uniformly and compare the antipodal vertices

$$x = \frac{\xi}{\sqrt{d}}, \quad x' = -x.$$

Let $L = 2Y - 1$ and

$$Z_i = \xi_i L.$$

Under parameter s , symmetry gives

$$\mathbb{E}_s Z_i = \rho_d(\alpha) s_i, \quad Z_i^2 = 1. \quad (110)$$

For n independent predetermined comparisons, set

$$\widehat{s}_i = \text{sign} \left(\sum_{t=1}^n Z_{t,i} \right).$$

Hoeffding's inequality and (109) imply

$$\mathbb{P}_s(\widehat{s}_i \neq s_i) \leq \exp \left\{ -c_F \frac{n}{D_{d,b,\beta}} \right\}.$$

Thus there is a predetermined design of length

$$n_0 \leq C_{\gamma,F} D_{d,b,\beta} = C_{\gamma,F} \left(d + \frac{d^2}{\beta^2 b^2} \right)$$

with $\mathbb{E}_s d_H(\hat{s}, s) \leq \gamma d/32$ uniformly in s . The candidate vertex $\hat{x} = \hat{s}/\sqrt{d}$ belongs to $\mathcal{X}$, and

$$\Delta_s(\hat{x}, \hat{x}) = b - \langle \theta_s, \hat{x} \rangle = \frac{2b}{d} d_H(\hat{s}, s).$$

Because the posterior Bayes action can only improve on this candidate,

$$\mathbb{E} r_{\mathcal{X}}(Q_{n_0}) \leq \frac{2b}{d} \mathbb{E} d_H(\hat{s}, s) \leq \frac{\gamma b}{16}.$$

Markov's inequality gives

$$\mathbb{P}\{r_{\mathcal{X}}(Q_{n_0}) \leq \gamma b\} \geq \frac{15}{16}.$$

Stop the design at its first hit. Because the prior mean of θ_s is zero, every predetermined comparison has prior expected gap exactly

$$\mathbb{E}_{Q_0} \Delta_s(x, x') = b.$$

First-hit truncation only decreases nonnegative regret. Hence

$$\text{SC}_b^{\text{pref}}(Q_0) \leq C_{\gamma, F} b^2 D_{d, b, \beta}.$$

One-bit lower bound. Consider an arbitrary stopped experiment, let $\mathbf{M} = \{r_{\mathcal{X}}(Q_N) \leq \gamma b\}$, and put $p = \mathbb{P}(\mathbf{M})$. Include its independent random seed U in the initial history. Conditional on U , the stopped binary label strings form a prefix-free set. The source-coding inequality therefore gives

$$I(s; \mathcal{H}_N) = I(s; Y_{1:N} \mid U) \leq (\log 2) \mathbb{E} N. \quad (111)$$

On the other hand, product structure of the uniform sign prior gives

$$\begin{aligned} I(s; \mathcal{H}_N) &= \mathbb{E} D_{\text{KL}}(Q_{\mathcal{H}_N} \| Q_0) \\ &\geq \sum_{i=1}^d \mathbb{E} \text{kl} \left(\frac{1 + m_i(\mathcal{H}_N)}{2} \parallel \frac{1}{2} \right) \geq \frac{1}{2} \mathbb{E} \sum_{i=1}^d m_i(\mathcal{H}_N)^2. \end{aligned} \quad (112)$$

If $\mathbf{M}$ occurs, then

$$(1 - \gamma)b \leq h_{\mathcal{X}}(\bar{\theta}_N) \leq \|\bar{\theta}_N\|_2 = \frac{b}{\sqrt{d}} \|m(\mathcal{H}_N)\|_2.$$

Thus $\sum_i m_i^2 \geq (1 - \gamma)^2 d$ on $\mathbf{M}$, and (111)–(112) imply

$$\mathbb{E} N \geq c_{\gamma} p d. \quad (113)$$

Sensitivity lower bound. Let $s^{(i)}$ denote s with coordinate i flipped, and let P_s^N be the stopped-transcript law. For a query x, x' , put

$$u = \beta \langle \theta_s, x - x' \rangle, \quad v = \beta \langle \theta_{s^{(i)}}, x - x' \rangle.$$

Then

$$|u - v| = \frac{2\beta b}{\sqrt{d}} |x_i - x'_i|.$$

For the logistic link,

$$D_{\text{KL}}(\text{Ber } \sigma(u) \| \text{Ber } \sigma(v)) \leq \frac{1}{8} (u - v)^2,$$

while for the probit link, Gaussian thresholding and data processing give

$$D_{\text{KL}}(\text{Ber } \Phi(u) \| \text{Ber } \Phi(v)) \leq \frac{1}{2}(u - v)^2.$$

The stopped chain rule and $\|x - x'\|_2^2 \leq 4$ therefore yield

$$2^{-d} \sum_s \sum_{i=1}^d D_{\text{KL}}(P_s^N \| P_{s^{(i)}}^N) \leq C_F \frac{\beta^2 b^2}{d} \mathbb{E}N. \quad (114)$$

For unbounded N , first apply the identity to $N \wedge L$ and pass to the limit; the claim is immediate if $\mathbb{E}N = \infty$.

For completeness, joint convexity gives the posterior lower bound behind the reverse inequality. If $P_{i,+}$ and $P_{i,-}$ are the transcript laws conditional on $s_i = +1$ and $s_i = -1$, respectively, then pairing s with $s^{(i)}$ and mixing the remaining coordinates shows

$$2^{-d} \sum_s D_{\text{KL}}(P_s^N \| P_{s^{(i)}}^N) \geq \frac{1}{2} \{D_{\text{KL}}(P_{i,+} \| P_{i,-}) + D_{\text{KL}}(P_{i,-} \| P_{i,+})\}.$$

Under $\bar{P} = (P_{i,+} + P_{i,-})/2$, the posterior mean is

$$m_i = \frac{dP_{i,+} - dP_{i,-}}{dP_{i,+} + dP_{i,-}}.$$

Since $\log((1+z)/(1-z)) \geq 2z$ for $z \in (-1, 1)$, the last Jeffreys divergence is at least

$$4\mathbb{E}_{\bar{P}} m_i^2.$$

Summing over i and restricting to M lower-bounds the left side of (114) by

$$2(1 - \gamma)^2 p d.$$

Consequently,

$$\mathbb{E}N \geq c_{\gamma, Fp} \frac{d^2}{\beta^2 b^2}. \quad (115)$$

Combining (113) and (115) gives

$$\mathbb{E}N \geq c_{\gamma, Fp} D_{d,b,\beta}.$$

Before the first hit, every query has posterior expected gap at least γb . Lemma 2.2 therefore gives

$$\frac{b \mathbb{E}R_N}{p} \geq \gamma b^2 \frac{\mathbb{E}N}{p} \geq c_{\gamma, F} b^2 D_{d,b,\beta},$$

proving the stopped lower bound.

Fixed-scale regret. Apply Theorem 2.3 at Q_0 , with a constant strictly below the stopped lower bound, to obtain

$$\mathfrak{R}_{d,b,\beta,\mathcal{X}}^{\text{pref}}(T) \geq c_{\gamma, F} b \min\{T, D_{d,b,\beta}\}.$$

For the reverse inequality, take

$$L_0 = \lceil C_F D_{d,b,\beta} \log(eT) \rceil.$$

If $L_0 \geq T$, play arbitrary pairs. If $L_0 < T$, run the corresponding predetermined design for L_0 queries and then play $(\hat{s}/\sqrt{d}, \hat{s}/\sqrt{d})$. The concentration bounds above and a union bound give $\mathbb{P}_s(\hat{s} \neq s) \leq T^{-2}$. Here $L_0 < T$ implies $d < T$, so $\log(eT)$ absorbs the union over coordinates. Exploration costs at most bL_0 , and exploitation costs at most $2bT$ on the failure event. This proves (59). $\square$

E Fresh-subspace lower bounds

Proof of Theorem 6.3. Put

$$q(x) = \varepsilon + \frac{1}{2}\|x\|^2.$$

For $B \in S_2^{d-1}(1)$, $r \in [0, 1]$, and $u = \langle \theta, B \rangle$,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2} \left(s - \sqrt{1 + r^2 - 2ru} \right)_+^2.$$

Minimizing the square root over $r \in [0, 1]$ shows that the supremum of the right side is $(s - 1)^2/2$ when $u \leq 0$, and

$$\frac{1}{2} \left(s - \sqrt{1 - u^2} \right)^2$$

when $u > 0$. On $|u| \leq 1/2$, its square is at most $C(\varepsilon^4 + u^8)$; the complementary spherical cap has exponentially small probability. Rotation invariance and

$$\mathbb{E}_B \langle \theta, B \rangle^8 = \frac{105}{d(d+2)(d+4)(d+6)}$$

therefore give, uniformly in θ ,

$$\mathbb{E}_B \sup_{0 \leq r \leq 1} |F_\theta(rB) - q(rB)|^2 \leq C(\varepsilon^4 + d^{-4}). \quad (116)$$

Now condition on an arbitrary posterior ν . The fresh Haar direction B is independent of Θ , and $R(B)$ is known after conditioning on B . Comparing the Gaussian observation channel with the Θ -independent reference $q(R(B)B) + \xi$, the information-radius bound gives

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq \frac{1}{2} \mathbb{E}_{\Theta \sim \nu, B} |F_\Theta(RB) - q(RB)|^2.$$

Equation (116) proves (70).

Under the spherical prior, for a fixed nonadaptive design point $x = ru$, rotation invariance interchanges a fixed direction u under uniform Θ with a uniform direction B under fixed θ . Thus (116) also gives

$$\mathbb{E}_\Theta |F_\Theta(x) - q(x)|^2 \leq C(\varepsilon^4 + d^{-4}).$$

Condition on a possibly randomized design and compare its joint Gaussian observation law with the product reference $(q(X_i) + \xi_i)_{i \leq n}$. Additivity of the reference KL divergence proves (71).

For a stopped fresh-direction experiment, let $N_n = N \wedge n$ and use the cemetery-padded transcript. Since $\{N \geq t\}$ is H_{t-1} -measurable, conditional application of (70) and the chain rule give

$$I(\Theta; H_{N_n}) \leq C(\varepsilon^4 + d^{-4}) \sum_{t=1}^n \mathbb{P}(N \geq t) = C(\varepsilon^4 + d^{-4}) \mathbb{E} N_n.$$

Continuity of relative entropy along the increasing stopped σ -fields, followed by $n \rightarrow \infty$, proves (72).

For the block-fresh extension, condition on a block-start history and write U for the new Haar m -subspace. For $z = \|P_U \theta\|^2$ and every $x \in U \cap B_2^d(1)$,

$$\|\theta - x\|^2 = 1 - z + \|P_U \theta - x\|^2 \geq 1 - z.$$

Consequently, with $D_\theta(x) = q(x) - F_\theta(x)$,

$$\begin{aligned} 0 \leq D_\theta(x) &\leq \frac{1}{2} (s - \sqrt{1-z})^2 \\ &= \frac{1}{2} \left(\frac{2\varepsilon + z}{s + \sqrt{1-z}} \right)^2, \quad D_\theta(x)^2 \leq C(\varepsilon^4 + z^4). \end{aligned} \quad (117)$$

For fixed θ , the projection has the beta-moment identity

$$\mathbb{E}_U \|P_U \theta\|^8 = \frac{m(m+2)(m+4)(m+6)}{d(d+2)(d+4)(d+6)}.$$

Under the hypothesis of Remark 6.4, the same conditional expectation is instead bounded by $\kappa(m/d)^4$, uniformly in θ . Compare the complete, adaptively stopped transcript of one block with the channel that uses the same query rule but replaces every observation mean by $q(X_t)$. Sequential Gaussian KL, the information-radius bound, (117), and the cap of ℓ queries give

$$I(\Theta; \text{block transcript} \mid H_{\text{pre}}) \leq C\ell A_{d,m,\varepsilon},$$

uniformly over the incoming posterior. This block-level comparison is important: the within-block duration may be longer in unusually informative subspaces.

Let K_N be the number of nonempty blocks started by time N . Cemetery-pad the sequence of block transcripts and apply the chain rule at block boundaries. Since $K_N \leq N$,

$$I(\Theta; H_N) \leq C\ell A_{d,m,\varepsilon} \mathbb{E}K_N \leq C\ell A_{d,m,\varepsilon} \mathbb{E}N,$$

first for bounded N and then by monotone passage. This proves (74).

For the converse, convexity at the origin and $\nabla F_\theta(0) = -b_0\theta$ give

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x \rangle\}. \quad (118)$$

With $m_N = \mathbb{E}[\Theta \mid H_N]$, taking the infimum over x in (118) gives directly

$$r(Q_{H_N}) \geq b_0(1 - \|m_N\|).$$

Hence $r(Q_{H_N}) \leq \gamma b_0$ on $\mathbf{M}$ implies $\|m_N\| \geq 1 - \gamma$. The uniform law σ on $S_2^{d-1}(1)$ is $1/(d-1)$ -sub-Gaussian. The variational formula for relative entropy therefore yields

$$D_{\text{KL}}(\nu \parallel \sigma) \geq \frac{d-1}{2} \|\mathbb{E}_\nu \Theta\|^2.$$

Apply this to each posterior and average on $\mathbf{M}$:

$$I(\Theta; H_N) = \mathbb{E}D_{\text{KL}}(Q_{H_N} \parallel \sigma) \geq \frac{d-1}{2} (1 - \gamma)^2 \mathbb{P}(\mathbf{M}).$$

Combining this inequality with (72) proves (76); combining it with (74) proves (77). $\square$

Proof of the lower bound in Corollary 6.6. Let $N = \tau_\gamma \wedge T$ and $p = \mathbb{P}(\tau_\gamma \leq T)$. Before the contraction time, every action has conditional expected regret at least $r(Q_{t-1}) > \gamma b_0$. Hence

$$\mathbb{E}R_T \geq \gamma b_0 \mathbb{E}N. \quad (119)$$

Moreover,

$$\mathbb{E}N \geq T(1 - p).$$

Apply Theorem 6.3 to the stopped transcript H_N . On $\{\tau_\gamma \leq T\}$, its terminal posterior risk is at most γb_0 , and therefore

$$Cd^{-4}\mathbb{E}N \geq I(\Theta; H_N) \geq c_\gamma d p$$

when $\varepsilon = d^{-1}$. Thus

$$\mathbb{E}N \geq c_\gamma d^5 p.$$

Minimizing $\max\{T(1-p), c_\gamma d^5 p\}$ over $p \in [0, 1]$ gives $\mathbb{E}N \geq c_\gamma \min\{T, d^5\}$. Since $b_0 \asymp d^{-1}$, (119) proves (84). $\square$

Proof of Corollary 6.7. Theorem 6.3 and Lemma 2.2 give, at every admissible ε ,

$$\text{SC}_{b_0}^{\text{bf}(m, \ell)} \geq c_\gamma \frac{b_0^2 d}{\ell A_{d, m, \varepsilon}}. \quad (120)$$

When $\varepsilon = m/d$, one has $b_0 \asymp m/d$ and

$$A_{d, m, \varepsilon} \leq C(m/d)^4.$$

This proves the restricted stopped-cost and query bounds in (86)–(87). Run Theorem 6.5 with $a = b_0$ and truncate at the first contraction. It succeeds with constant probability, uses $\tilde{O}_\gamma(d^2/b_0^2) = \tilde{O}_\gamma(d^4/m^2)$ queries, and has regret $\tilde{O}_\gamma(d^2/b_0)$. Multiplication by b_0 proves $\text{SC}_{b_0} \leq \tilde{O}_\gamma(d^2)$.

For a T -round block-fresh policy, let $N = \tau_\gamma \wedge T$ and $p = \mathbb{P}(\tau_\gamma \leq T)$. As in (119),

$$\mathbb{E}R_T \geq \gamma b_0 \mathbb{E}N, \quad \mathbb{E}N \geq T(1-p), \quad \mathbb{E}N \geq c_\gamma \frac{d^5}{\ell m^4} p.$$

Balancing the last two inequalities and using $b_0 \asymp m/d$ proves (88).

Finally take $T_\star \asymp d^5/(\ell m^4)$. If $\ell m^2 \leq d^{1-\delta}$, then the target

$$a \asymp \frac{d}{\sqrt{T_\star}}$$

lies below b_0 , and the polynomial slack absorbs the logarithmic enlargement used in Theorem 6.5. The same explore–then–commit calculation as in the proof of (85) gives unrestricted regret $\tilde{O}_\gamma(d\sqrt{T_\star})$, which is the displayed $\tilde{O}_\gamma(d^{7/2}/(\sqrt{\ell} m^2))$. Dividing the restricted and unrestricted quantities gives the stated gaps. $\square$

F The dimple regret bounds

Proof of the upper statements in Corollary 6.6. The same adaptive experiment, run to accuracy a and followed by commitment, gives the family upper bound. At $a = b_0 \asymp \varepsilon$, Theorem 6.5 uses $\tilde{O}(d^2/\varepsilon^2)$ queries and $\tilde{O}(d^2/\varepsilon)$ regret. Equation (82) proves the adaptive part of the query separation. Since $\varepsilon^4 + d^{-4} \leq 2\varepsilon^4$ in the stated range, Theorem 6.3 proves the lower part for nonadaptive experiments under the spherical prior and for fresh-direction experiments. Specializing to $\varepsilon = d^{-1}$ gives the displayed endpoint.

For the regret upper bound, first note that playing the origin for all T rounds costs $b_0 T \leq CT/d$. When $T \geq C_\gamma d^4 L_{d, T}^2$, choose

$$a = A_\gamma \frac{d L_{d, T}}{\sqrt{T}}$$

with A_γ large enough. Then $a \leq b_0$, $L_{d,a} \leq CL_{d,T}$, and (80) uses at most $T/2$ rounds. Indeed, enlarging the crossover constant makes $a \leq c/d \leq b_0$, while $1/a \leq \sqrt{T}/d$ and $\log(d/a) \leq \log(dT)$ give the displayed logarithmic comparison directly from (78). The exploration regret is at most

$$\frac{C_\gamma d^2 L_{d,a}^2}{a} \leq C_\gamma d L_{d,T} \sqrt{T}.$$

Play $\hat{x}$ for the remaining rounds. Since losses are bounded by one, (79) gives

$$\sup_{\theta} \mathbb{E}_{\theta} F_{\theta}(\hat{x}) \leq (\gamma_0 + c_0)a \leq C_\gamma a.$$

The exploitation regret is therefore at most $C_\gamma a T \leq C_\gamma d L_{d,T} \sqrt{T}$. Below the displayed crossover, use the origin strategy. If $T \leq d^4$, its cost CT/d is the smaller term in (85). If $d^4 < T < C_\gamma d^4 L_{d,T}^2$, then

$$\frac{T}{d} \leq C_\gamma L_{d,T} d \sqrt{T}.$$

These two cases prove (85) throughout the remaining range. $\square$

G Proof of the observation-adaptive experiment

Throughout this appendix, reduce ε_0 , if necessary, so that $\varepsilon_0 \leq 1/8$. Put

$$s_+ = \sqrt{1 + 2\varepsilon_0}, \quad \beta_- = \frac{2}{s_+ + 1}.$$

Then, uniformly over the parameter range,

$$1 \leq s \leq s_+, \quad \beta_- \varepsilon \leq b_0 = s - 1 \leq \varepsilon. \quad (121)$$

Fix

$$\gamma_0 = \frac{\gamma}{4}, \quad c_\delta = \frac{\gamma}{4}, \quad c_0 = c_\delta, \quad \delta = c_\delta a,$$

so that $\gamma_0 + c_0 = \gamma/2 < \gamma$, and put

$$\mathcal{L} = 1 + \log\{16d(1 + \log(d/a))/\delta\}.$$

Thus $\mathcal{L} \leq C_\gamma L_{d,a}$ after increasing $C_L(\gamma)$.

We now record the order in which the remaining constants are chosen. First choose a constant A_H large enough for all the Haar-coordinate bounds below. Next fix $C_{\text{geo}} \geq 4$, and then choose

$$0 < q_\star \leq \min \left\{ \frac{1}{8}, \frac{1}{4C_{\text{geo}}} \right\}. \quad (122)$$

In particular,

$$q_\star < s/4, \quad C_{\text{geo}} q_\star \leq s/2$$

uniformly in ε . The initialization step c_h and its repetition constant are then chosen; this produces the numerical correlation constant $\kappa_{\text{init}} > 0$ in (131) below. After κ_{init} is fixed, choose c_2 ; this fixes the geometric constants B_λ, c_4, k_0 . Next choose k and $\rho = \sqrt{1 + k^2}$, and only then choose the amplification repetition constants C_{gate} and C_{amp} , whose simultaneous confidence bounds depend on the resulting deterministic iteration cap. Finally choose c_5 , then c_{term} , then c_6 , and last C_{con} .

The compatible inequalities imposed on these constants are displayed at the point where each is used.

We use the following conditional form of sphere concentration. If $(u_i)_{i=1}^k$ is a Haar orthonormal basis of a fixed k -dimensional subspace and w is a fixed unit vector in that subspace, then, after increasing A_H ,

$$\max_{1 \leq i \leq k} |\langle w, u_i \rangle| \leq A_H \sqrt{\mathcal{L}/k} \quad (123)$$

except on an event of probability at most $\exp(-4\mathcal{L})$, provided $\mathcal{L} \leq k$. When $\mathcal{L} > k$, we instead use the deterministic bound 1.

Every basis is drawn conditionally on the preceding transcript. Under $\mathbb{P}_\theta$, the current iterate and the tangent component of the fixed vector θ are then fixed, so (123) applies conditionally after normalizing a nonzero tangent component. Every empirical mean uses fresh Gaussian noise. The initialization, amplification, and contraction stages contain at most $C(1 + \log(d/a))$ concentration calls in total. The definition of $\mathcal{L}$, together with sufficiently large repetition constants, therefore gives events $\mathcal{H}_{\text{init}}, \mathcal{H}_{\text{amp}}, \mathcal{H}_{\text{con}}$ such that, uniformly in θ ,

$$\mathbb{P}_\theta(\mathcal{H}_{\text{init}}^c) \leq \delta/3, \quad \mathbb{P}_\theta(\mathcal{H}_{\text{amp}}^c \mid \mathcal{H}_{\text{init}}) \leq \delta/3, \quad \mathbb{P}_\theta(\mathcal{H}_{\text{con}}^c \mid \mathcal{H}_{\text{init}} \cap \mathcal{H}_{\text{amp}}) \leq \delta/3.$$

Write $\mathcal{H} = \mathcal{H}_{\text{init}} \cap \mathcal{H}_{\text{amp}} \cap \mathcal{H}_{\text{con}}$. All three stages have deterministic caps. Off $\mathcal{H}$, the same capped routine is continued, returning the origin whenever a normalization is undefined and taking the stopping branch at every gate equality. These conventions make the complete experiment total and Borel measurable.

Whenever $r = \|\theta - x\| < s$, expanding (67) gives

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x \rangle - 1. \quad (124)$$

G.1 Orthogonal initialization

Lemma G.1. *With probability at least $1 - \delta/3$, the construction below returns a unit vector v_0 satisfying*

$$\langle \theta, v_0 \rangle \geq c \min \left\{ 1, \max \left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d} \right) \right\}. \quad (125)$$

It uses at most $Cd^2\mathcal{L}^2/\varepsilon^2$ queries and incurs at most $Cd^2\mathcal{L}^2/\varepsilon$ regret on the success event.

Proof. Draw a Haar basis $(u_i)_{i=1}^d$, and set

$$h_0 = c_h \min \{ \sqrt{\varepsilon}, \varepsilon \sqrt{d/\mathcal{L}} \}, \quad n_0 = \lceil C_{\text{init}} d \mathcal{L}^2 / \varepsilon^2 \rceil. \quad (126)$$

Estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using n_0 repetitions at each of the two points. Let A_0 be a universal constant for the Taylor estimate used below. If $\mathcal{L} \leq d$, then $h_0 = c_h \varepsilon / t$, where

$$t = \max \{ \sqrt{\varepsilon}, \sqrt{\mathcal{L}/d} \},$$

and (123) gives

$$h_0^2 + 2h_0 \max_i |\langle \theta, u_i \rangle| \leq (c_h^2 + 2A_H c_h) \varepsilon.$$

If $\mathcal{L} > d$, then $h_0 = c_h \varepsilon \sqrt{d/\mathcal{L}}$, and the deterministic coordinate bound gives the same inequality with A_H replaced by one. Choose $c_h \leq 1/8$ so that

$$c_h^2 + 2 \max\{A_H, 1\} c_h \leq 1, \quad A_0 c_h^2 \leq \frac{\beta_-}{8}. \quad (127)$$

Both query points are then strictly in the active dimple. Equation (124) gives

$$D_i^0 = -\lambda_i \langle \theta, u_i \rangle, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \quad |\lambda_i - b_0| \leq A_0 h_0^2, \quad (128)$$

where $r_i^\pm = \|\theta \mp h_0 u_i\|$. Hence the noiseless difference vector is

$$D = -b_0 \theta + e_0, \quad \|e_0\| \leq A_0 h_0^2 \leq \frac{\beta_-}{8} \varepsilon.$$

Write $\widehat{D} = D + \zeta$. Conditional on the basis, ζ is an isotropic Gaussian vector with coordinate variance $(2n_0 h_0^2)^{-1}$. On $\mathcal{H}_{\text{init}}$,

$$|\langle \theta, \zeta \rangle| \leq \frac{C\varepsilon}{\sqrt{C_{\text{init}}} h_0 \sqrt{d\mathcal{L}}}, \quad (129)$$

$$\|\zeta\| \leq \frac{C\varepsilon}{\sqrt{C_{\text{init}}} h_0 \mathcal{L}} \left(1 + \sqrt{\mathcal{L}/d}\right). \quad (130)$$

Set

$$t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}.$$

If $\mathcal{L} \leq d$, then $h_0 = c_h \varepsilon / t$, and

$$\frac{t}{\sqrt{d\mathcal{L}}} \leq \varepsilon, \quad \frac{t^2}{\mathcal{L}} \left(1 + \sqrt{\mathcal{L}/d}\right) \leq 2\varepsilon.$$

These inequalities follow by considering separately $t^2 = \varepsilon$ and $t^2 = \mathcal{L}/d$. If $\mathcal{L} > d$, then $t = 1$, $h_0 = c_h \varepsilon \sqrt{d/\mathcal{L}}$, and

$$\frac{1}{d} \leq \varepsilon, \quad \frac{1 + \sqrt{\mathcal{L}/d}}{\sqrt{d\mathcal{L}}} \leq \frac{2}{d} \leq 2\varepsilon.$$

Consequently, after choosing C_{init} sufficiently large,

$$|\langle \theta, \zeta \rangle| \leq \frac{\beta_-}{8} \varepsilon, \quad \|\zeta\| \leq B_{\text{init}} \frac{\varepsilon}{t}$$

for a fixed universal B_{init} . Together with (121) and (127), this yields

$$\langle \theta, -\widehat{D} \rangle \geq \frac{\beta_-}{2} \varepsilon, \quad \|\widehat{D}\| \leq (2 + B_{\text{init}}) \frac{\varepsilon}{t}.$$

Thus $\widehat{D} \neq 0$, and, with

$$\kappa_{\text{init}} = \frac{\beta_-}{2(2 + B_{\text{init}})}, \quad c_{\text{init}} := \kappa_{\text{init}} t,$$

the vector $v_0 = -\widehat{D}/\|\widehat{D}\|$ satisfies

$$\langle \theta, v_0 \rangle \geq c_{\text{init}} = \kappa_{\text{init}} \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}. \quad (131)$$

This implies (125).

The stage uses $2dn_0 = O(d^2 \mathcal{L}^2 / \varepsilon^2)$ queries. Every query point has loss $O(\varepsilon)$, so its regret is $O(d^2 \mathcal{L}^2 / \varepsilon)$. $\square$

G.2 Tangent amplification

Lemma G.2. *Conditionally on the initialization event, a capped tangent routine reaches $\|\theta - v\| \leq q_\star$ with failure probability at most $\delta/3$. With c_{init} defined in (131), the stage uses*

$$N_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^4}, \quad R_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2}, \quad (132)$$

up to lower-order $C\mathcal{L}(1 + \log(1/c_{\text{init}}))$ gate costs.

Proof. Let

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s - q)_+^2, \quad \Delta_\star = \phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2).$$

Because $q_\star < s$,

$$\Delta_\star = \frac{sq_\star}{2} - \frac{3q_\star^2}{8} \geq \frac{q_\star}{2} - \frac{3q_\star^2}{8} > 0. \quad (133)$$

The lower bound is universal.

The routine maintains a unit vector v_j and a deterministic number $\underline{c}_j$ satisfying

$$\langle \theta, v_j \rangle \geq \underline{c}_j, \quad \underline{c}_j = \rho^j c_{\text{init}}, \quad (134)$$

where $\rho > 1$ is fixed below. If $\underline{c}_j > 1 - q_\star^2/8$, then

$$\|\theta - v_j\|^2 = 2\{1 - \langle \theta, v_j \rangle\} < q_\star^2/4,$$

and the routine stops.

Otherwise, average $m_{\text{gate}} = \lceil C_{\text{gate}}\mathcal{L} \rceil$ observations at v_j , and compare the average with

$$T_{\text{gate}} = \frac{\phi_\varepsilon(q_\star/2) + \phi_\varepsilon(q_\star)}{2}.$$

The routine stops when the average is at most T_{gate} , and continues otherwise. After k and ρ are fixed below, choose C_{gate} large enough so that, simultaneously at every gate,

$$|\bar{Y}_j - F_\theta(v_j)| \leq \Delta_\star/4$$

on $\mathcal{H}_{\text{amp}}$. Since $F_\theta(v_j) = \phi_\varepsilon(\|\theta - v_j\|)$ and ϕ_ε is nondecreasing, stopping implies

$$\|\theta - v_j\| \leq q_\star, \quad (135)$$

whereas continuing implies

$$\|\theta - v_j\| \geq q_\star/2. \quad (136)$$

Suppose the gate continues. Write

$$c = \underline{c}_j, \quad v = v_j, \quad p = \langle \theta, v \rangle, \quad x = (c/2)v, \quad r_0 = \|\theta - x\|.$$

Then $p \geq c$, and

$$r_0^2 = 1 + \frac{c^2}{4} - cp \leq 1 - \frac{3c^2}{4}, \quad r_0 \geq 1 - \frac{c}{2} \geq \frac{1}{2}. \quad (137)$$

$$c^2 \geq c_{\text{init}}^2 \geq \kappa_{\text{init}}^2 \varepsilon. \quad (138)$$

Draw a Haar orthonormal basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$, and write $q_i = \langle \theta, u_i \rangle$. When $\mathcal{L} \leq d-1$, set $h = c_2 c$. On the conditional Haar event,

$$|q_i| \leq A_H \sqrt{\mathcal{L}/(d-1)} \leq \frac{2A_H}{\kappa_{\text{init}}} c.$$

When $\mathcal{L} > d-1$, set $h = c_2 c^2$ and use $|q_i| \leq 1$; in this case $\mathcal{L}/d \geq 1/2$, so $c \geq c_{\text{init}} \geq \kappa_{\text{init}}/\sqrt{2}$. Choose $c_2 \leq 1/8$ so that

$$c_2^2 + \frac{4A_H}{\kappa_{\text{init}}} c_2 \leq \frac{3}{8}, \quad c_2^2 + 2c_2 \leq \frac{3}{8}. \quad (139)$$

In the two cases, respectively,

$$h^2 + 2h|q_i| \leq \frac{3c^2}{8}.$$

Together with (137), this shows that $x \pm hu_i$ lie in the active dimple. Their norms are at most one after decreasing c_2 , if necessary.

For $r_i^\pm = \|\theta - x \mp hu_i\|$, the exact centered difference is

$$D_i = \frac{F_\theta(x + hu_i) - F_\theta(x - hu_i)}{2h} = -\lambda_i q_i, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \quad (140)$$

Let $\lambda_0 = s/r_0 - 1$. From (137), for a universal constant $B_0 < \infty$,

$$\lambda_0 \geq 1 - r_0 \geq \frac{3c^2}{8}, \quad \lambda_0 \leq B_0 c, \quad (141)$$

where the upper bound uses $s - 1 \leq \varepsilon \leq c^2/\kappa_{\text{init}}^2$ and $r_0 \geq 1/2$. A Taylor expansion of $\sqrt{r_0^2 + z}$, uniform over the preceding activity margin, gives

$$|\lambda_i - \lambda_0| \leq A_\lambda h^2$$

for a universal A_λ . Decrease c_2 further so that

$$A_\lambda c_2^2 \leq \frac{3}{16}. \quad (142)$$

Since $c \leq 1$, both choices of h then imply

$$\frac{3c^2}{16} \leq \lambda_i \leq B_\lambda c \quad (143)$$

for a universal B_λ .

Set

$$g = -\sum_{i=1}^{d-1} D_i u_i = \sum_{i=1}^{d-1} \lambda_i q_i u_i.$$

If $q = \|\theta - v\|$, then $p = 1 - q^2/2$, and (136) gives

$$\sum_i q_i^2 = 1 - p^2 = q^2(1 - q^2/4) \geq q_\star^2/8.$$

Therefore (143) implies

$$\langle \theta, g \rangle \geq c_4 c^2, \quad \|g\| \leq B_\lambda c, \quad c_4 := \frac{3q_\star^2}{128}. \quad (144)$$

Estimate every D_i with

$$n_c = \lceil C_{\text{amp}} d\mathcal{L}/c^4 \rceil$$

repetitions at each of $x + hu_i$ and $x - hu_i$. With empirical differences $\widehat{D}_i$, define

$$\widehat{g} = - \sum_{i=1}^{d-1} \widehat{D}_i u_i = g + \zeta.$$

Conditional on the basis, ζ is isotropic in $v^\perp$, with coordinate variance $(2n_ch^2)^{-1}$. For $\mathcal{L} \leq d-1$, this variance is at most

$$\frac{Cc^2}{C_{\text{amp}}d\mathcal{L}},$$

for $\mathcal{L} > d-1$, it is at most

$$\frac{C}{C_{\text{amp}}d\mathcal{L}},$$

and c is bounded below by a universal multiple of κ_{init} . After k and ρ are fixed below, choose C_{amp} large enough so that, simultaneously at all stages,

$$\|\zeta\| \leq c, \quad |\langle \theta, \zeta \rangle| \leq \frac{c_4}{2} c^2 \quad (145)$$

on $\mathcal{H}_{\text{amp}}$.

It follows that $\widehat{g} \neq 0$, and for $u = \widehat{g}/\|\widehat{g}\|$,

$$\langle \theta, u \rangle \geq k_0 c, \quad k_0 := \frac{c_4}{2(B_\lambda + 1)}.$$

Choose

$$0 < k \leq \min\{k_0/2, q_\star/4\}, \quad \rho = \sqrt{1+k^2} > 1, \quad (146)$$

and set

$$v_{j+1} = \frac{v_j + ku}{\sqrt{1+k^2}}, \quad \underline{c}_{j+1} = \rho \underline{c}_j.$$

Because $u \perp v_j$,

$$\langle \theta, v_{j+1} \rangle \geq \frac{c + kk_0 c}{\sqrt{1+k^2}} \geq \sqrt{1+k^2} c = \underline{c}_{j+1}.$$

Thus (134) is preserved. The restriction $k \leq q_\star/4$ also ensures $\rho(1 - q_\star^2/8) < 1$, so the deterministic lower bounds never exceed one before the next stopping check.

After at most

$$1 + \left\lceil \frac{\log\{(1 - q_\star^2/8)/c_{\text{init}}\}}{\log \rho} \right\rceil = O(1 + \log(1/c_{\text{init}}))$$

iterations, either a gate has stopped or the deterministic correlation threshold has been crossed. In both cases the returned vector satisfies $\|\theta - v\| \leq q_\star$.

At correlation level c , the tangent queries number $O(d^2\mathcal{L}/c^4)$. By (138),

$$F_\theta(x \pm hu_i) \leq \varepsilon + \frac{1}{2} \|x \pm hu_i\|^2 \leq Cc^2,$$

so their regret is $O(d^2\mathcal{L}/c^2)$. The gate uses $O(\mathcal{L})$ queries and regret per iteration. Geometric summation over $c_j = \rho^j c_{\text{init}}$ gives

$$N_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^4} + C\mathcal{L}(1 + \log(1/c_{\text{init}})),$$

and

$$R_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2} + C\mathcal{L}(1 + \log(1/c_{\text{init}})).$$

Finally,

$$c_{\text{init}} \geq \kappa_{\text{init}} \max\{\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\}$$

unless it is already a fixed constant. This yields (132) and the simplified bounds

$$O(d^2\mathcal{L}/\varepsilon^2) \quad \text{and} \quad O(d^2\mathcal{L}/\varepsilon),$$

respectively. □

G.3 Angular contraction

Lemma G.3. *Conditionally on the success events of the first two stages, the capped contraction stage returns $\hat{x}$ satisfying $F_\theta(\hat{x}) \leq \gamma_0 a$ with conditional probability at least $1 - \delta/3$. It has deterministic query cap*

$$C_\gamma d^2 \mathcal{L}/a^2$$

and success-event regret at most

$$C_\gamma d^2 \mathcal{L}/a.$$

Proof. Choose

$$0 < c_5 \leq \frac{1}{8}, \quad 0 < c_{\text{term}} \leq \frac{\gamma_0}{s_+}. \quad (147)$$

Let

$$Q_j = q_\star 2^{-j}, \quad J = \min\{j \geq 0 : Q_j \leq c_{\text{term}} a\}.$$

The routine maintains the invariant

$$\|v_j\| = 1, \quad \|\theta - v_j\| \leq Q_j. \quad (148)$$

The amplification lemma establishes it at $j = 0$. For each $j < J$, write $v = v_j$, $Q = Q_j$, set $h = c_5 Q$, and take a measurably chosen orthonormal basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$. Define

$$\alpha_h = (1 + h^2)^{-1/2}, \quad t = h(1 + h^2)^{-1/2}, \quad x_i^\pm = \alpha_h v \pm t u_i.$$

Then $x_i^\pm$ are unit vectors, $t \geq c_5 Q/\sqrt{2}$, and

$$\|x_i^\pm - v\|^2 = 2(1 - \alpha_h) \leq h^2.$$

Consequently,

$$\|\theta - x_i^\pm\| \leq (1 + c_5)Q \leq C_{\text{geo}}Q \leq C_{\text{geo}}q_\star \leq s/2. \quad (149)$$

Thus every probe lies in the active dimple.

Write

$$\theta = c_v v + w, \quad c_v = \langle \theta, v \rangle = 1 - \frac{1}{2}\|\theta - v\|^2, \quad w = \sum_i q_i u_i.$$

Since $Q \leq q_\star \leq 1/8$, one has $c_v > 0$,

$$c_v = \sqrt{1 - \|w\|^2}, \quad \|w\| \leq \|\theta - v\| \leq Q < 1/2. \quad (150)$$

Let

$$\psi(r) = sr - \frac{1}{2}r^2, \quad G(p) = \psi(\sqrt{2-2p}).$$

Define

$$I_Q = \{p \in [-1, 1] : \sqrt{2-2p} \leq C_{\text{geo}}Q\}.$$

By (149), the inner products

$$p_i^\pm = \langle \theta, x_i^\pm \rangle = \alpha_h c_v \pm tq_i$$

belong to I_Q , and $F_\theta(x_i^\pm) = G(p_i^\pm)$. The restriction $G_Q := G|_{I_Q}$ is strictly decreasing. Moreover, for $p \in I_Q$,

$$|(G_Q^{-1})'(G(p))| = \frac{\sqrt{2-2p}}{s - \sqrt{2-2p}} \leq 2C_{\text{geo}}Q. \quad (151)$$

The right side tends to zero at $p = 1$, so this estimate includes the endpoint by continuity. Set

$$B_q := \frac{2\sqrt{2}C_{\text{geo}}}{c_5},$$

and choose $c_6 > 0$ so that

$$2B_q c_6 \leq \frac{1}{2}. \quad (152)$$

Let $\bar{Y}_i^\pm$ be the sample means at $x_i^\pm$, and let $\tilde{Y}_i^\pm$ be their Euclidean projections onto the interval $G(I_Q)$. Define

$$\hat{q}_i = \frac{G_Q^{-1}(\tilde{Y}_i^+) - G_Q^{-1}(\tilde{Y}_i^-)}{2t}, \quad \hat{w} = \sum_i \hat{q}_i u_i.$$

Use

$$n_Q = \lceil C_{\text{con}} d \mathcal{L} / Q^2 \rceil$$

repetitions at each site. Choose C_{con} so that, simultaneously for all i and all $j < J$,

$$|\bar{Y}_i^\pm - F_\theta(x_i^\pm)| \leq \frac{c_6 Q}{\sqrt{d}} \quad (153)$$

on $\mathcal{H}_{\text{con}}$. Projection onto $G(I_Q)$ cannot increase the error because the true value belongs to this interval. Combining (151), (153), and $t \geq c_5 Q / \sqrt{2}$ gives

$$|\hat{q}_i - q_i| \leq B_q \frac{c_6 Q}{\sqrt{d}},$$

and hence

$$\|\hat{w} - w\| \leq B_q c_6 Q. \quad (154)$$

Let $\tilde{w}$ be the Euclidean projection of $\hat{w}$ onto the ball of radius $1/2$. Since the true w belongs to this ball by (150),

$$\|\tilde{w} - w\| \leq \|\hat{w} - w\|.$$

Set

$$v_{j+1} = \sqrt{1 - \|\tilde{w}\|^2} v + \tilde{w}.$$

The map

$$z \mapsto \sqrt{1 - \|z\|^2} v + z$$

is 2-Lipschitz on the ball of radius $1/2$. Using $\theta = \sqrt{1 - \|w\|^2} v + w$, we obtain

$$\|\theta - v_{j+1}\| \leq 2\|\tilde{w} - w\| \leq Q/2 = Q_{j+1}.$$

This proves (148) by induction.

The number of queries at scale Q_j is $O(d^2 \mathcal{L}/Q_j^2)$. By (149),

$$F_\theta(x_i^\pm) = \psi(\|\theta - x_i^\pm\|) \leq s_+ C_{\text{geo}} Q_j,$$

so the corresponding regret is $O(d^2 \mathcal{L}/Q_j)$. If $J = 0$, this stage makes no queries. Otherwise, Q_j decreases geometrically and $Q_{J-1} > c_{\text{term}} a$, so

$$\sum_{j < J} \frac{d^2 \mathcal{L}}{Q_j^2} \leq \frac{C d^2 \mathcal{L}}{c_{\text{term}}^2 a^2}, \quad \sum_{j < J} \frac{d^2 \mathcal{L}}{Q_j} \leq \frac{C d^2 \mathcal{L}}{c_{\text{term}} a}.$$

The output $\hat{x} = v_J$ then satisfies

$$F_\theta(\hat{x}) = \psi(\|\theta - v_J\|) \leq s_+ \|\theta - v_J\| \leq s_+ c_{\text{term}} a \leq \gamma_0 a.$$

The number J and every repetition count are deterministic. Continuing the same clipped updates off $\mathcal{H}_{\text{con}}$ therefore gives the claimed deterministic cap. $\square$

Proof of Theorem 6.5. Run Lemmas G.1–G.3 in order, assigning failure probability $\delta/3$ to each. The three conditional failure bounds imply $\sup_\theta \mathbb{P}_\theta(\mathcal{H}^c) \leq \delta = c_0 a$. The amplification lower bounds and the angular invariant are asserted only on $\mathcal{H}$; the deterministic caps, clipping rules, and fixed iteration limits define the same routine on $\mathcal{H}^c$. Because $a \leq b_0 \asymp \varepsilon$, their deterministic caps and geometric sums give

$$N \leq C_\gamma d^2 L_{d,a}^2 / a^2$$

and, on $\mathcal{H}$,

$$R_N \leq C_\gamma d^2 L_{d,a}^2 / a.$$

On $\mathcal{H}^c$, the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since $\mathbb{P}(\mathcal{H}^c) \leq \delta = c_\delta a$, this adds at most

$$C_\gamma d^2 L_{d,a}^2 / a$$

to expected regret. The accuracy statement follows from Lemma G.3 and the union bound. Since losses are bounded by one, the same statement gives, under every prior,

$$\mathbb{E} F_\Theta(\hat{x}) \leq (\gamma_0 + c_0) a.$$

Apply Markov's inequality to the nonnegative random variable $\mathbb{E}[F_\Theta(\hat{x}) \mid H_N]$ to obtain (82). On the displayed event, the posterior Bayes risk is no larger than the posterior loss of $\hat{x}$, and hence is at most γa . $\square$

H Gaussian models and Bayes–minimax equality

This appendix proves Theorem 5.1. It establishes measurable selection of statewise experiments and finite-horizon Bayes–minimax equality; Section 2 already proved the two-sided stopped-cost conversion.

Proof of Corollary 5.3. The upper endpoint is the current general BCO upper bound (Lattimore and Györfy, 2023; Fokkema et al., 2024); the lower endpoint is the minimax theorem of Rajaraman (2026). Its 1-Lipschitz hard family is on the Euclidean unit ball and is anchored by $f(0) = 0$. Consequently $f(x) \in [-1, 1]$, so it embeds in the present bounded unit-noise model by the affine rescaling $g = (f + 1)/2 \in [0, 1]$. From an observation $f(x) + \xi$, return $(f(x) + \xi + 1)/2 + \zeta$, where $\zeta \sim N(0, 3/4)$ is independent. The result is $g(x) + \xi'$ with $\xi' \sim N(0, 1)$, and regret changes only by a factor two. Apply Corollary 5.2. $\square$

Lemma H.1 (Prior-wise Borel realization). *Fix a Borel prior Q and a finite horizon T . On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy inducing the same joint law of (F, H_T) under Q . The same holds when the action space is augmented by a stopping symbol.*

Proof. At time t , represent the action kernel as a universally measurable map from histories into the standard Borel space of action distributions. Under the reached history law, this map has a Borel version. Replace it by that version; the strategic law through time t is unchanged. Induction over $t = 1, \dots, T$ proves the claim. Adding one isolated stopping symbol preserves the standard Borel property. $\square$

H.1 Finite-horizon Bayes–minimax identity

Let $\nu = N(0, 1)$, let $H_T = (C \times \mathbb{R})^T$, and equip $\mathcal{P}(H_T)$ with the topology of weak convergence. Let $\mathcal{P}_T$ be the set of reference strategic measures

$$P(dh_T) = \prod_{t=1}^T \pi_t(dx_t \mid h_{t-1}) \nu(dy_t)$$

generated by Borel randomized nonanticipating policies.

Lemma H.2 (Compactness of reference strategic measures). *Suppose that C is a nonempty compact metric space. Then $\mathcal{P}_T$ is a nonempty convex weakly compact subset of $\mathcal{P}(H_T)$.*

More precisely, a probability measure P on H_T belongs to $\mathcal{P}_T$ if and only if, for every $t \leq T$, $a \in C_b((C \times \mathbb{R})^{t-1} \times C)$, and $b \in C_b(\mathbb{R})$,

$$\int a(h_{t-1}, x_t) b(y_t) dP = \left(\int b d\nu \right) \int a(h_{t-1}, x_t) dP \quad (155)$$

Proof. Every reference strategic measure satisfies (155) by iterated conditioning. Conversely, the bounded-continuous identities extend, by a measure-determining and monotone-class argument on Polish spaces, to bounded Borel a and b . They therefore imply

$$P(Y_t \in \cdot \mid H_{t-1}, X_t) = \nu(\cdot) \quad P\text{-a.s.}$$

Because all spaces are standard Borel, P has Borel regular conditional distributions

$$\pi_t(\cdot \mid h_{t-1}) = P(X_t \in \cdot \mid H_{t-1} = h_{t-1}).$$

Successive disintegration, together with the displayed conditional law of Y_t , gives

$$P(dh_T) = \prod_{t=1}^T \pi_t(dx_t \mid h_{t-1}) \nu(dy_t),$$

so $P \in \mathcal{P}_T$. This proves the characterization.

The characterization is linear in P , so $\mathcal{P}_T$ is convex. It also implies that the $Y_{1:T}$ -marginal of every $P \in \mathcal{P}_T$ is $\nu^{\otimes T}$. Since C^T is compact, this common observation marginal makes $\mathcal{P}_T$ tight.

If $P_n \in \mathcal{P}_T$ and $P_n \Rightarrow P$, then both sides of (155) converge for every bounded continuous a, b . Hence P satisfies the same identities and, by the characterization, belongs to $\mathcal{P}_T$. Thus $\mathcal{P}_T$ is weakly closed. Prokhorov's theorem now gives weak compactness. $\square$

For $f \in \mathcal{C}(C, [0, 1])$, write

$$m(f) = \min_{x \in C} f(x), \quad G_f(h_T) = \sum_{t=1}^T \{f(x_t) - m(f)\},$$

and

$$L_f(h_T) = \exp \left\{ \sum_{t=1}^T \left(y_t f(x_t) - \frac{1}{2} f(x_t)^2 \right) \right\},$$

and define

$$u(P, f) = \int L_f(h_T) G_f(h_T) P(dh_T).$$

Lemma H.3 (Continuity of the Gaussian regret payoff). *The map*

$$(P, f) \mapsto u(P, f)$$

is jointly continuous on $\mathcal{P}_T \times \mathcal{C}(C, [0, 1])$, where the first space has the weak topology and the second has the uniform topology. It is affine in P and takes values in $[0, T]$.

If P is the reference strategic measure generated by a policy π , then

$$L_f = \frac{dP_f^\pi}{dP} \quad \text{on } \mathbf{H}_T, \quad u(P, f) = \mathbb{E}_f^\pi R_T.$$

Proof. The likelihood-ratio identity follows by multiplying the one-step density ratios

$$\frac{\varphi(y_t - f(x_t))}{\varphi(y_t)} = \exp \left\{ y_t f(x_t) - \frac{1}{2} f(x_t)^2 \right\}.$$

The action kernels cancel because the same policy is used under the reference and shifted observation laws. Thus $L_f P$ is the actual strategic law, and

$$u(P, f) = \mathbb{E}_f^\pi R_T \in [0, T].$$

Fix f . The integrand $L_f G_f$ is continuous on $\mathbf{H}_T$, but it is unbounded. Put

$$S(y_{1:T}) = \sum_{t=1}^T |y_t|.$$

Since $0 \leq G_f \leq T$,

$$0 \leq L_f(h_T) G_f(h_T) \leq T e^{S(y_{1:T})}. \quad (156)$$

The right-hand side is integrable under $\nu^{\otimes T}$, and every $P \in \mathcal{P}_T$ has this same observation marginal. Let $\chi_M \in C_c(\mathbb{R}^T)$ satisfy $0 \leq \chi_M \leq 1$ and $\chi_M(y) \rightarrow 1$ pointwise. Then $\chi_M(y_{1:T}) L_f(h_T) G_f(h_T)$ is bounded and continuous, while

$$\sup_{P \in \mathcal{P}_T} \int (1 - \chi_M) L_f G_f dP \leq T \int (1 - \chi_M(y)) e^{S(y)} \nu^{\otimes T}(dy) \rightarrow 0.$$

It follows that weak convergence $P_n \Rightarrow P$ implies $u(P_n, f) \rightarrow u(P, f)$.

It remains to control changes in f , uniformly over P . Let $\delta = \|f - f'\|_\infty$. Since

$$|m(f) - m(f')| \leq \delta, \quad |G_f - G_{f'}| \leq 2T\delta,$$

and, upon writing $A_f = \log L_f$,

$$|A_f - A_{f'}| \leq \delta\{S(y_{1:T}) + T\}, \quad A_f, A_{f'} \leq S(y_{1:T}),$$

the mean-value theorem gives

$$|L_f G_f - L_{f'} G_{f'}| \leq T\delta e^{S(y_{1:T})} \{S(y_{1:T}) + T + 2\}.$$

Therefore

$$\sup_{P \in \mathcal{P}_T} |u(P, f) - u(P, f')| \leq K_T \|f - f'\|_\infty,$$

where

$$K_T = T \int e^{S(y)} \{S(y) + T + 2\} \nu^{\otimes T}(dy) < \infty.$$

Combining this uniform estimate with continuity in P for fixed f proves joint continuity. Affinity in P is immediate. $\square$

Proof of (63). Fix $C \in \mathcal{C}_d$ and abbreviate $\mathfrak{F} = \mathfrak{F}_d(C)$. Let $\mathcal{V}_{\text{fs}}$ be the real vector space of finitely supported signed measures on $\mathfrak{F}$, equipped with the locally convex topology generated by the seminorms

$$q \mapsto \left| \int \phi(f) q(df) \right|, \quad \phi \in C_b(\mathfrak{F}).$$

Let $\mathcal{Q}_0 \subset \mathcal{V}_{\text{fs}}$ be the convex set of finitely supported probability measures, and define

$$U(P, Q) = \int_{\mathfrak{F}} u(P, f) Q(df).$$

By Lemma H.3, $f \mapsto u(P, f)$ is bounded and continuous. Hence $Q \mapsto U(P, Q)$ is continuous and affine on $\mathcal{Q}_0$. For each $Q \in \mathcal{Q}_0$, $P \mapsto U(P, Q)$ is a finite convex combination of continuous affine functions and is therefore continuous and affine. Lemma H.2 makes $\mathcal{P}_T$ compact and convex. Sion's theorem (Sion, 1958) gives

$$\min_{P \in \mathcal{P}_T} \sup_{Q \in \mathcal{Q}_0} U(P, Q) = \sup_{Q \in \mathcal{Q}_0} \min_{P \in \mathcal{P}_T} U(P, Q).$$

Because every Dirac measure belongs to $\mathcal{Q}_0$,

$$\sup_{Q \in \mathcal{Q}_0} U(P, Q) = \sup_{f \in \mathfrak{F}} u(P, f).$$

The preceding lemmas identify every $P \in \mathcal{P}_T$ with a Borel randomized nonanticipating policy and identify $u(P, f)$ with that policy's regret under f . Consequently,

$$\inf_{\pi} \sup_{f \in \mathfrak{F}} \mathbb{E}_f^\pi R_T = \sup_{Q \in \mathcal{Q}_0} \inf_{\pi} \mathbb{E}_Q^\pi R_T. \quad (157)$$

For an arbitrary Borel prior Q on $\mathfrak{F}$, if $P_n \Rightarrow P$, Lemma H.3 gives $u(P_n, f) \rightarrow u(P, f)$ for every f . Since $0 \leq u \leq T$, dominated convergence makes $P \mapsto \int u(P, f) Q(df)$ continuous. Compactness of $\mathcal{P}_T$ therefore attains the minimum, and

$$\min_{P \in \mathcal{P}_T} \int u(P, f) Q(df) \leq \min_{P \in \mathcal{P}_T} \sup_{f \in \mathfrak{F}} u(P, f).$$

Taking the supremum over all Borel priors cannot exceed the minimax value, whereas, by (157), the supremum over finitely supported priors already equals the minimax value. Hence

$$\inf_{\pi} \sup_{f \in \mathfrak{F}} \mathbb{E}_f^{\pi} R_T = \sup_{Q \text{ on } \mathfrak{F}} \inf_{\pi} \mathbb{E}_Q^{\pi} R_T,$$

and the supremum on the right may be restricted to finitely supported priors. Taking the outer supremum over $C \in \mathcal{C}_d$ proves (63). $\square$

H.2 Measurable selection of stopped experiments

Lemma H.4 (Capped Gaussian selection). *Fix a query cap $L < \infty$. The capped stopped value*

$$v_L(Q, b) = \inf_{\mathcal{E}: N \leq L} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}$$

is universally measurable. For every $\epsilon > 0$, there is a universally measurable state-to-experiment rule with value at most $v_L(Q, b) + \epsilon$ at every active state.

Proof. The belief space $\mathcal{P}(\mathfrak{F}_d(C))$ is standard Borel. For a belief μ , define

$$\ell(\mu, x) = \int \{f(x) - \min_C f\} \mu(df), \quad r(\mu) = \min_{x \in C} \ell(\mu, x).$$

The first map is Borel and continuous in x ; compactness of C makes r Borel and supplies a Borel Bayes-action selector. With φ the standard Gaussian density, the posterior update is the Borel map

$$\Psi(\mu, x, o)(df) = \frac{\varphi(o - f(x)) \mu(df)}{\int \varphi(o - f'(x)) \mu(df')}.$$

For a rational price $q > 0$, the capped penalized problem is exactly the recursion (7), with $\mu' = \Psi(\mu, x, o)$. Its one-step costs are bounded below and all primitive kernels are Borel. Standard integration, partial minimization, and universally measurable selection for lower-semianalytic dynamic programs therefore apply (Bertsekas and Shreve, 1978, Propositions 7.47–7.50). In particular, its root value $J_L^q(\mu, b)$ is lower semianalytic and has universally measurable near-minimizers.

Directly from the definition,

$$v_L(\mu, b) < q \iff J_L^q(\mu, b) < 0.$$

The rational strict sublevel sets make v_L universally measurable. On $\{v_L < +\infty\}$, enumerate the positive rationals and measurably choose the first q with

$$v_L + \epsilon/4 < q < v_L + \epsilon/2.$$

Then $J_L^q < 0$. Partition this set further by the first integer n such that $J_L^q \leq -1/n$, and use a Bellman selector whose approximation error is less than $1/(2n)$. The resulting penalized value remains negative, so its success probability is positive and its ratio is below $q < v_L + \epsilon/2$. On $\{v_L = +\infty\}$, any measurable one-query rule satisfies the extended-value approximation. Countable patching gives one universally measurable ϵ -optimal rule. $\square$

Lemma H.5 (Removing the query cap). *The unrestricted value $\text{SC}_b(Q)$ is universally measurable and has a universally measurable ϵ -optimal selector for every $\epsilon > 0$. The selected experiment may have a finite deterministic cap $L(Q, b)$ depending on the state.*

Proof. Truncate an admissible experiment after L queries. As $L \rightarrow \infty$, its stopped regret and hit probability converge monotonically to those of the complete experiment:

$$\mathbb{E}R_{N \wedge L} \uparrow \mathbb{E}R_N, \quad \mathbb{P}(\mathbf{M}_L) \uparrow \mathbb{P}(\mathbf{M}).$$

Hence its ratio converges, including the zero-success case under the $+\infty$ convention. Indeed, if $\mathbb{P}(\mathbf{M}) > 0$, componentwise convergence gives convergence of the ratios; if $\mathbb{P}(\mathbf{M}) = 0$, then every truncated hit probability is zero and all ratios are $+\infty$. Since capped experiments are a subset of all experiments, $\text{SC}_b(Q) \leq v_L(Q, b)$ for every L . Conversely, truncating any admissible experiment and passing to the limit gives the reverse inequality. Therefore

$$\text{SC}_b(Q) = \inf_{L \geq 1} v_L(Q, b).$$

This is universally measurable by Lemma H.4. On $\{\text{SC}_b < +\infty\}$, choose measurably the first L for which $v_L < \text{SC}_b + \epsilon/2$, then use the capped $\epsilon/2$ -selector. On $\{\text{SC}_b = +\infty\}$, use any measurable one-query rule; the approximation statement is vacuous there. Countable patching proves the claim. $\square$

Proof of (64). Lemma H.5 selects, simultaneously at every active state, an experiment within ϵ of its pointwise infimum. Taking the worst state and then letting $\epsilon \downarrow 0$ gives $\text{SC}^{\text{unif}} \leq \text{SC}^{\text{pt}}$. The reverse inequality is immediate from the definitions. For each fixed prior and finite global horizon, Lemma H.1 replaces the universal selector by a Borel policy with the same strategic law. $\square$

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